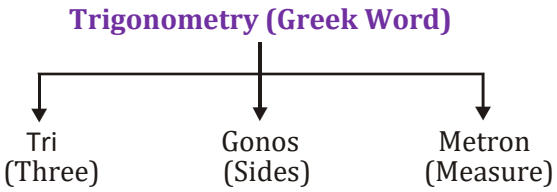
01 Trigonometric Ratios and Identities

INTRODUCTION:



The word 'trigonometry' is derived from the Greek words 'trigon' and 'metron' and it means 'measuring the sides of a triangle'. The subject was originally developed to solve geometric problems involving triangles. It was studied by sea captains for navigation, surveyor to map out the new lands, by engineers and others. Currently, trigonometry is used in many areas such as the science of seismology, designing electric circuits, describing the state of an atom, predicting the heights of tides in the ocean, analyzing a musical tone and in many other areas.

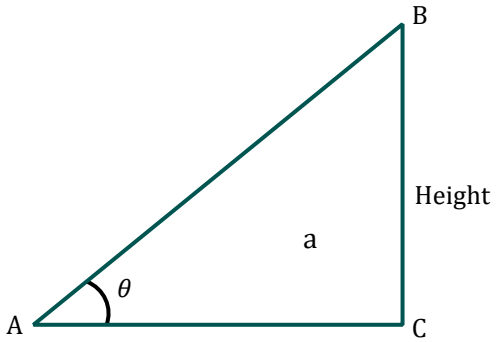
Trigonometry is the branch of mathematics in which we study about triangle. Basically there are six parameters in a triangle (Three sides & three angles).

How they are related to each other is mainly discussed in trigonometry.

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TRIGONOMETRY TO MEASURE HEIGHT OF TOWER BUILDING OR MOUNTAIN:

Trigonometry is used to in measuring the height of a building or a mountain. The distance of a building from the viewpoint and the elevation angle can easily determine the height of a building using the trigonometric functions.



It is used to find the distance of the shore from a point in the sea

It is also used in oceanography in calculating the height of tides in oceans.

TRIGONOMETRY IN AVIATION:

Aviation technology has been evolved in many up-gradations in the last few years. It has taken into account the speed, direction, and distance as well as have to consider the speed and direction of the wind. The wind plays a vital role in when and how a flight will travel. This equation is solved using trigonometry.

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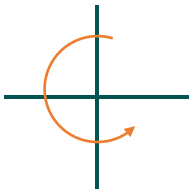
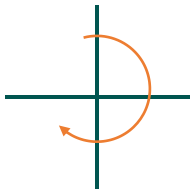
TRIGONOMETRY IN CRIMINOLOGY:

Trigonometry is even used in the investigation of a crime scene. The functions of trigonometry are helpful to calculate a trajectory of a projectile and to estimate the causes of a collision in a car accident. Further, it is used to identify how an object falls or in what angle the gun is shot.

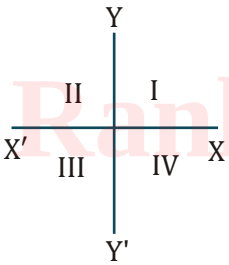
TRIGONOMETRY IN MARINE BIOLOGY:

Trigonometry is often used by marine biologists for measurements to figure out depth of sunlight that affects algae to photosynthesis. Using the trigonometric function and mathematical models, marine biologists estimate the size of larger animals like whales and also understand their behaviors.

MEASUREMENT OF ANGLE:

Sign convention:  Anticlockwise direction will indicate +ve angle,  Clockwise turning will indicate –ve angle

If α is the angle measured from positive x-axis in anticlockwise direction.



$0° < \alpha < 90°$ I[st] quadrant

$90° < \alpha < 180°$ II[nd] quadrant

$180° < \alpha < 270°$ III[rd] quadrant

$270° < \alpha < 360°$ IV[th] quadrant

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SYSTEM OF MEASUREMENT OF ANGLE:

System of Measurement of angle

Degree (English) (Sexagesimal system	Radian (Circular system)	French (Grade system) (Centisimal system)

Degree system : When we draw a line perpendicular to another and then divide the angle between them in 90 equal parts. Then each part will be equal to one degree.

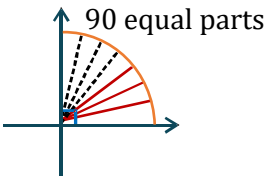
$1° = 60' \ \& \ 1' = 60''$

Hence 90° = 1 right angle

or 180° = 2 right angle.

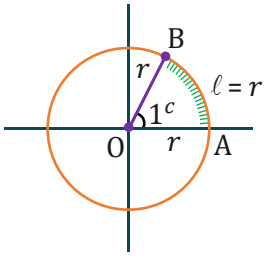
Grade system : One right angle = 100^g grades, $1^g = 100'$ minutes, $1' = 100''$ seconds

Radian : If the length of arc (ℓ) is equal to r (radius), then the angle subtended by it at the center of the circle is equal to 1 radian. (1^c)



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Commonly, $\boxed{\text{Angle} = \frac{\text{arc}}{\text{radius}}}$, i.e. $\ell = r\theta$, where θ is the angle subtended at the center by circular arc of length ℓ. If $\ell = r$ then $\theta = 1^c$



If $\ell = 2r$ then $\theta = 2^c$

If $\ell = 3r$ then $\theta = 3^c$

$\vdots$

If $\ell = kr$ then $\theta = k^c$

$\vdots$

If $\ell = \pi r$ then $\theta = \pi^c$

Hence $\pi^c = 180°$ or $\frac{\pi^c}{2} = 90° \Rightarrow$ 1 right angle $= \frac{\pi}{2}$ radians

$\Rightarrow 1° = \frac{\pi^c}{180} = 0.01746$ rad & $1^c = \frac{180^o}{\pi} = \frac{180}{22/7} = \frac{630}{11} = 57.2°$

Relation in three systems : $\frac{D}{90} = \frac{G}{100} = \frac{R}{\pi/2}$

Illustration 1

Find the radian measures corresponding to the following degree measures:

(i) 25° **(ii)** – 47° 30′

Solution:

(i) 25°

Here 180° = π radians

It can be written as

$25° = \frac{\pi}{180} \times 25$ radian

So we get

$= \frac{5\pi}{36}$ radian

(ii) – 47°30′

Here 1° = 60′

It can be written as

$-47°30' = -47\frac{1}{2}$ degree

So we get

$= \frac{-95}{2}$ degree

Here 180° = π radian

$= \frac{-95}{2}$ degree $= \frac{\pi}{180} \times \left(\frac{-95}{2}\right)$ radian

It can be written as

$= \left(\frac{-19}{36\times2}\right)\pi$ radian $= \frac{-19}{72}\pi$ radian

We get

$-47°30' = \frac{-19}{72}\pi$ radian

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Illustration 2

Find the degree measures corresponding to the following radian measures (Use π = 22/7)

(i) 11/16 **(ii)** –4

Solution:

(i) 11/16

Here $\frac{1}{2}$ radian = 180°

$\frac{11}{16}$ radian = $\frac{180}{\pi} \times \frac{11}{16}$ degree

We can write it as

$= \frac{45 \times 11}{\pi \times 4}$ degree

Se we get

$= \frac{45 \times 11 \times 7}{22 \times 4}$ degree

$= \frac{315}{8}$ degree

(ii) – 4

Here π radian = 180°

– 4 radian = $\frac{180}{\pi} \times (-4)$ degree

We can write it as

$= \frac{180 \times 7(-4)}{22}$ degree

By further calculation

$= \frac{-2520}{11}$ degree

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Illustration 3

If the arcs of same length in two circles subtend angles of 60° and 75° at their centers. Find the ratio of their radii.

Solution:

Let r_1 and r_2 be the radii of the given circles and let their arcs of same length 's' subtend angles of 60° and 75° at their centers.

Now, $60° = \left(60 \times \frac{\pi}{180}\right)^c = \left(\frac{\pi}{3}\right)$ and $75° = \left(75 \times \frac{\pi}{180}\right)^c = \left(\frac{5\pi}{12}\right)^c$

$\therefore \frac{\pi}{3} = \frac{s}{r_1}$ and $\frac{5\pi}{12} = \frac{s}{r_2}$

$\Rightarrow \frac{\pi}{3} r_1 = s$ and $\frac{5\pi}{12} r_2 = s$

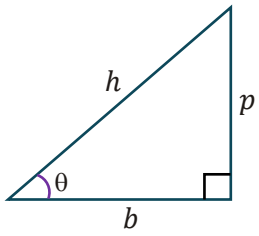
$\Rightarrow \frac{\pi}{3} r_1 = \frac{5\pi}{12} r_2$

$\Rightarrow 4r_1 = 5r_2$

$\Rightarrow r_1 : r_2 = 5 : 4$

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T-RATIOS (OR TRIGONOMETRIC FUNCTIONS):



In a right angle triangle

$\sin\theta = \frac{p}{h}$; $\cos\theta = \frac{b}{h}$; $\tan\theta = \frac{p}{b}$; $\operatorname{cosec}\theta = \frac{h}{p}$; $\sec\theta = \frac{h}{b}$ and $\cot\theta = \frac{b}{p}$

'p' is perpendicular; 'b' is base and 'h' is hypotenuse.

BASIC TRIGONOMETRIC IDENTITIES:

(1) $\sin\theta.\operatorname{cosec}\theta = 1$

(2) $\cos\theta.\sec\theta = 1$

(3) $\tan\theta.\cot\theta = 1$

(4) $\tan\theta = \frac{\sin\theta}{\cos\theta}$ & $\cot\theta = \frac{\cos\theta}{\sin\theta}$

(5) $\sin^2\theta + \cos^2\theta = 1$ or $\sin^2\theta = 1 - \cos^2\theta$ or $\cos^2\theta = 1 - \sin^2\theta$

(6) $\sec^2\theta - \tan^2\theta = 1$ or $\sec^2\theta = 1 + \tan^2\theta$ or $\tan^2\theta = \sec^2\theta - 1$

(7) $\sec\theta + \tan\theta = \frac{1}{\sec\theta - \tan\theta}$

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(8) $\operatorname{cosec}^2\theta - \cot^2\theta = 1$ or $\operatorname{cosec}^2\theta = 1 + \cot^2\theta$ or $\cot^2\theta = \operatorname{cosec}^2\theta - 1$

(9) $\operatorname{cosec}\theta + \cot\theta = \frac{1}{\operatorname{cosec}\theta - \cot\theta}$

Values of T-Ratios of Some Standard Angles:

Angles / T-ratio	0°	30°	45°	60°	90°
	0	π/6	π/4	π/3	π/2
sinθ	0	1/2	$1/\sqrt{2}$	$\sqrt{3}/2$	1
cosθ	1	$\sqrt{3}/2$	$1/\sqrt{2}$	½	0
tanθ	0	$1/\sqrt{3}$	1	$\sqrt{3}$	N.D.
cotθ	N.D.	$\sqrt{3}$	1	$1/\sqrt{3}$	0
secθ	1	$2/\sqrt{3}$	$\sqrt{2}$	2	N.D.
cosecθ	N.D.	2	$\sqrt{2}$	$2/\sqrt{3}$	1

Illustration 4

If $\sin\theta + \sin^2\theta = 1$, then prove that $\cos^{12}\theta + 3\cos^{10}\theta + 3\cos^8\theta + \cos^6\theta - 1 = 0$

Solution:

Given that $\sin\theta = 1 - \sin^2\theta = \cos^2\theta$

L.H.S. $= \cos^6\theta(\cos^2\theta + 1)^3 - 1 = \sin^3\theta(1 + \sin\theta)^3 - 1 = (\sin\theta + \sin^2\theta)^3 - 1 = 1 - 1 = 0$

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Illustration 5

$4(\sin^6\theta + \cos^6\theta) - 6(\sin^4\theta + \cos^4\theta)$ is equal to

(A) 0 (B) 1 (C) –2 (D) none of these

Ans. (C)

Solution:

$4[(\sin^2\theta + \cos^2\theta)^3 - 3\sin^2\theta\cos^2\theta(\sin^2\theta + \cos^2\theta)] - 6[(\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta]$

$= 4[1 - 3\sin^2\theta\cos^2\theta] - 6[1 - 2\sin^2\theta\cos^2\theta]$

$= 4 - 12\sin^2\theta\cos^2\theta - 6 + 12\sin^2\theta\cos^2\theta = -2$

Illustration 6

Prove that : (i) $\sin^4\theta + \cos^4\theta = 1 - 2\sin^2\theta\cos^2\theta$ (ii) $\sin^6\theta + \cos^6\theta = 1 - 3\sin^2\theta\cos^2\theta$.

Solution:

(i) $\sin^4\theta+\cos^4\theta = (\sin^2\theta+\cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta = 1 - 2\sin^2\theta\cos^2\theta$

(ii) $\sin^6\theta+\cos^6\theta = (\sin^2\theta)^3+(\cos^2\theta)^3$

$= (\sin^2\theta+\cos^2\theta)^3 - 3\sin^2\theta\cos^2\theta(\sin^2\theta+\cos^2\theta)$ $(\because (a+b)^3 = a^3 + b^3 + 3ab(a+b))$

$= 1 - 3\sin^2\theta\cos^2\theta$

Illustration 7

Prove that $\sqrt{\frac{1-\cos\theta}{1+\cos\theta}} = \operatorname{cosec}\theta - \cot\theta$

Solution:

$$\sqrt{\frac{1-\cos\theta}{1+\cos\theta}} = \sqrt{\frac{(1-\cos\theta)(1-\cos\theta)}{(1+\cos\theta)(1+\cos\theta)}} = \frac{1-\cos\theta}{\sin\theta} = \operatorname{cosec}\theta - \cot\theta$$

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Illustration 8

Prove that $\sqrt{\sec^2 A + \operatorname{cosec}^2 A} = \tan A + \cot A$, when $0° < A < 90°$

Solution:

$$\sqrt{\sec^2 A + \operatorname{cosec}^2 A} = \sqrt{\frac{1}{\cos^2 A} + \frac{1}{\sin^2 A}} = \frac{1}{\sin A\cos A} = \frac{\sin^2 A+\cos^2 A}{\sin A\cos A} = \tan A + \cot A$$

OR $\sqrt{1 + \tan^2 A + 1 + \cot^2 A} = \sqrt{\tan^2 A + \cot^2 A + 2(\tan A.\cot A)} = \tan A + \cot A$

Illustration 9

$$\frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1} =$$

(A) $\frac{1-\sin\theta}{\cos\theta}$ (B) $\frac{1-\cos\theta}{\sin\theta}$ (C) $\frac{1+\sin\theta}{\cos\theta}$ (D) $\frac{1+\cos\theta}{\sin\theta}$

Ans. (C)

Solution:

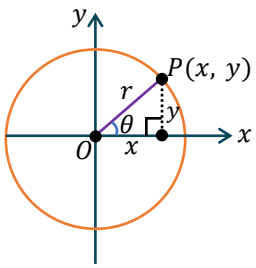
$$\frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1} = \frac{(\tan\theta+\sec\theta)-(\sec^2\theta-\tan^2\theta)}{\tan\theta-\sec\theta+1} \quad [\because \sec^2\theta - \tan^2\theta = 1]$$

$$= \frac{(\sec\theta+\tan\theta)\{1-(\sec\theta-\tan\theta)\}}{\tan\theta-\sec\theta+1} = \frac{(\sec\theta+\tan\theta)(\tan\theta-\sec\theta+1)}{\tan\theta-\sec\theta+1}$$

$$= \sec\theta + \tan\theta = \frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta} \Rightarrow \frac{1+\sin\theta}{\cos\theta}$$

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REAL DEFINITION OF SINE & COSINE OF A REAL NUMBER: (CIRCULAR FUNCTIONS):



$$\sin\theta = \frac{\text{distance of point P from } x\text{-axis}}{\text{radius of the circle}}$$

$$\Rightarrow \sin\theta = \frac{y}{r}$$

$$\text{Similarly, } \cos\theta = \frac{\text{distance of point } P \text{ from } y\text{-axis}}{\text{radius of the circle}}$$

$$\Rightarrow \cos\theta = \frac{x}{r}$$

As $\theta \to 0, y$ decreases, x increases,

$y \to 0, x \to r,$

$\therefore \sin 0 = 0$ & $\cos 0 = 1$

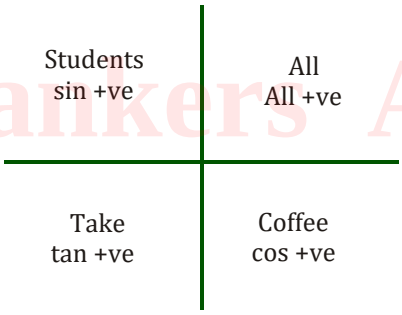
As $\theta \to 90°, y$ increases, x decreases

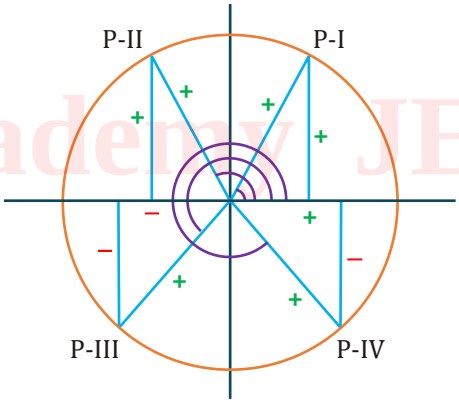
$y \to r, x \to 0$

$\therefore \sin 90° = 1$ & $\cos 90° = 0$

Note : All other trigonometric ratios can be derived using sine and cos definition.

SIGN OF T-RATIOS:





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Illustration 10

If $\cos x = -\frac{3}{5}$, x lies in the third quadrant, find the values of other five trigonometric functions.

Solution:

Since $\cos x = -\frac{3}{5}$, we have $\sec x = -\frac{5}{3}$

Now, $\sin^2 x + \cos^2 x = 1$, i.e., $\sin^2 x = 1 - \cos^2 x$

or $\sin^2 x = 1 - \frac{9}{25} = \frac{16}{25}$

Hence $\sin x = \pm\frac{4}{5}$

Since x lies in third quadrant, $\sin x$ is negative. Therefore

$\sin x = -\frac{4}{5}$

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which also gives

$\text{cosec}\, x = -\frac{5}{4}$

Further, we have

$\tan x = \frac{\sin x}{\cos x} = \frac{4}{3}$ and $\cot x = \frac{\cos x}{\sin x} = \frac{3}{4}$

Illustration 11

If $\cot x = -\frac{5}{12}$, x lies in second quadrant, find the values of other five trigonometric functions.

Solution:

Since $\cot x = -\frac{5}{12}$, we have $\tan x = -\frac{12}{5}$

Now $\sec^2 x = 1 + \tan^2 x = 1 + \frac{144}{25} = \frac{169}{25}$

Hence $\sec x = \pm\frac{13}{5}$

Since x lies in second quadrant, $\sec x$ will be negative. Therefore

$\sec x = -\frac{13}{5}$,

which also gives

$\cos x = -\frac{5}{13}$

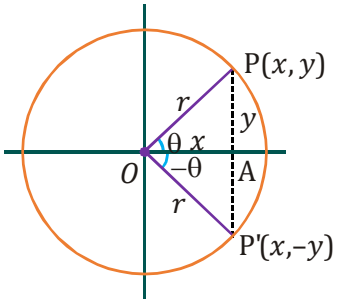
Further, we have

$\sin x = \tan x \cos x = \left(-\frac{12}{5}\right) \times \left(-\frac{5}{13}\right) = \frac{12}{13}$

and $\text{cosec}\, x = \frac{1}{\sin x} = \frac{13}{12}$

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TRIGONOMETRIC RATIOS OF ALLIED ANGLES:



(i) T-Ratios of Angle (–θ):

In ΔAOP & $\Delta AOP'$, we have $PA = P'A$

Hence $\angle POA = \angle P'OA$

$\therefore$ If $\angle POA$ measured anticlockwise is θ, then $\angle P'OA$ measured clockwise is (–θ)

$\therefore \sin(-\theta) = \frac{AP'}{OP} = -\frac{AP}{OP} = -\sin\theta$

$\cos(-\theta) = \frac{OA}{OP'} = \frac{OA}{OP} = \cos\theta$

$\tan(-\theta) = \frac{AP'}{OA} = -\frac{AP}{OA} = -\tan\theta$

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(ii) T-Ratios of Angle (90° – θ):

In Ist quadrant all trigonometric ratios are positive

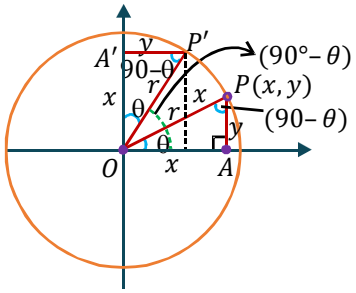
$\sin\theta = \frac{y}{r}, \cos\theta = \frac{x}{r}$.

ΔPOA & $\Delta P'OA'$ are congruent

$\therefore \sin(90° - \theta) = \frac{x}{r} = \cos\theta$

$\cos(90° - \theta) = \frac{y}{r} = \sin\theta$

$$\Rightarrow \begin{cases} \sin(90° - \theta) = \cos\theta & ; \quad \cot(90° - \theta) = \tan\theta \\ \cos(90° - \theta) = \sin\theta & ; \quad \operatorname{cosec}(90° - \theta) = \sec\theta \\ \tan(90° - \theta) = \cot\theta & ; \quad \sec(90° - \theta) = \operatorname{cosec}\theta \end{cases}$$



Note: θ and 90° – θ are called complementary angles.

Ex. sin60° = cos30°, tan10° = cot80°

(iii) T-Ratios of Angle (90° + θ):

$\sin(90° + \theta) = \frac{x}{r} = \cos\theta$

$\cos(90° + \theta) = \frac{-y}{r} = -\sin\theta$

tan(90°+ θ) = – cotθ replace θ by – θ in (i)

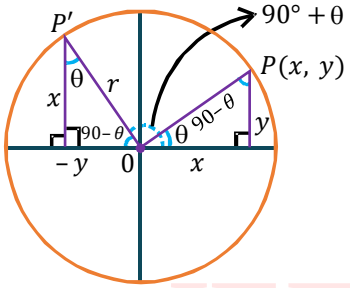
sin(90°–(–θ)) = sin(90° + θ) = cosθ

Ex. $\sin(120°) = \sin(90° + 30°) = \cos(30°) = \frac{\sqrt{3}}{2}$

cos(120°) = –sin(30°) = –1/2

tan(135°) = tan(90° + 45°) = –cot45° = –1

$\cos(150°) = \cos(90° + 60°) = -\sin 60° = -\frac{\sqrt{3}}{2}$



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(iv) T-Ratios of Angle (180° – θ):

$\sin(180° - \theta) = \frac{y}{r} = \sin\theta$

OR Replace θ by 90° – θ in sin(90° + θ) = cos θ

$\cos(180° - \theta) = \frac{-x}{r} = -\cos\theta$

tan(180° – θ) = – tanθ

Ex. $\sin120° = \sin(180° - 60°) = \sin60° = \frac{\sqrt{3}}{2}$

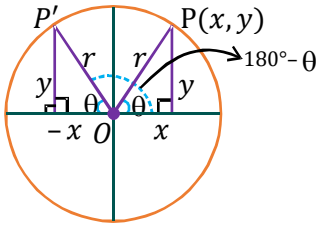


Illustration 12

cos 10° + cos 20° + cos 30° + ... + cos 80° + cos 100° + cos 150° + cos 160° + cos 170° = 0.

Solution:

cos10° + cos20° + cos30° +.....+ cos170°

We know that sum of cosine of supplementary angles is zero.

If α + β = 180° ; α = 180° – β ⇒ cosα = –cos β

cosα + cosβ = 0

cos10° + cos170° + cos20° + cos160° +.....+ cos80° + cos10° + cos90°= 0

Note: Similarly, if α + β = 180° then tanα + tanβ = 0

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Illustration 13

$\tan\frac{\pi}{11} + \tan\frac{2\pi}{11} + \tan\frac{4\pi}{11} + \tan\frac{7\pi}{11} + \tan\frac{9\pi}{11} + \tan\frac{10\pi}{11} = 0$

Solution:

$\tan\frac{\pi}{11} + \tan\frac{2\pi}{11} + \tan\frac{4\pi}{7} + \tan\frac{7\pi}{11} + \tan\frac{9\pi}{11} + \tan\frac{10\pi}{11}$

$\left(\tan\frac{\pi}{11} + \tan\frac{10\pi}{11}\right) + \left(\tan\frac{2\pi}{11} + \tan\frac{9\pi}{11}\right) + \tan\left(\frac{4\pi}{7} + \tan\frac{7\pi}{11}\right)$

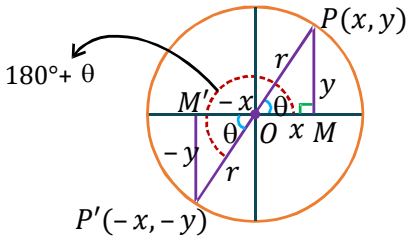
We know that if $\alpha + \beta = \pi$

$\tan\alpha + \tan\beta = 0$

$\therefore \left(\tan\frac{\pi}{11} + \tan\frac{10\pi}{11}\right) + \left(\tan\frac{2\pi}{11} + \tan\frac{9\pi}{11}\right) + \tan\left(\frac{4\pi}{7} + \tan\frac{7\pi}{11}\right) = 0$

(v) T-Ratios of Angle (180° + θ):

$\sin(180° + \theta) = \frac{-y}{r} = -\sin\theta$



OR Replace θ by 90° + θ in $\sin(90° + \theta) = \cos\theta$

$\cos(180° + \theta) = -\cos\theta$,

$\tan(180° + \theta) = \tan\theta$

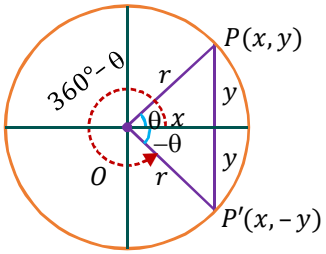
Ex. $\sin(210°) = \sin(180° + 30°) = -\sin 30° = \frac{-1}{2}$

$\cos(240°) = \cos(180° + 60°) = -\cos 60° = -\frac{1}{2}$

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(vi) T-Ratios of Angle (360° – θ):

$\sin(360° - \theta) = \frac{-y}{r} = -\sin\theta = \sin(-\theta)$



Replace θ by 270° – θ in $\sin(90° + \theta) = \cos\theta$

$\cos(360° - \theta) = \cos(-\theta) = \frac{x}{r} = \cos\theta$

$\tan(360° - \theta) = \tan(-\theta) = -\tan\theta$

Ex. $\sin(315°) = \sin(360° - 45°) = -\sin 45° = -\frac{1}{\sqrt{2}}$

$\cos(315°) = \cos 45° = \frac{1}{\sqrt{2}}$

$\tan(330°) = \tan(360° - 30°) = -\tan 30° = -\frac{1}{\sqrt{3}}$

$\tan(-120°) = -\tan 120° = -\tan(180° - 60°) = \tan 60° = \sqrt{3}$

MEMORY TIPS:

Rule-1:

(i) Reduction about 0° ± θ, 180° ± θ; 360° ± θ. No change in trigonometric ratio (sine remain sine)

(ii) Reduction about 90° ± θ, 270° ± θ. Change trigonometric ratio to its complementary. (sine change to cos)

For More Join: @IITJEE_Advanced

Rule-2:

Depending on the quadrant in which given angle lies, decide the sign as per trigonometric ratio being reduced.

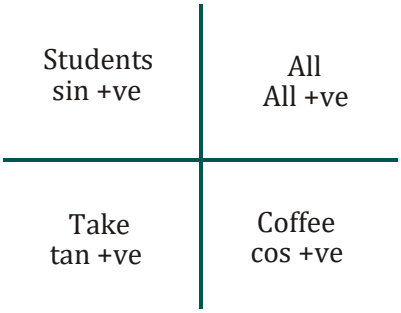


Illustration 14

Find the value of $\sin\frac{31\pi}{3}$.

Solution:

We know that values of $\sin x$ repeats after an interval of 2π. Therefore

$$\sin\frac{31\pi}{3} = \sin\left(10\pi + \frac{\pi}{3}\right) = \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}.$$

Illustration 15

Find the value of $\cos(-1710°)$.

Solution:

We know that values of $\cos x$ repeats after an interval of 2π or $360°$.

Therefore

$\cos(-1710°) = \cos(-1710° + 5 \times 360°)$

$= \cos(-1710° + 1800°) = \cos 90° = 0.$

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Illustration 16

Find the values of the trigonometric functions in Qus. 1 to 5.

1. $\sin 765°$ **2.** $\operatorname{cosec}(-1410°)$ **3.** $\tan\frac{19\pi}{3}$

4. $\sin\left(-\frac{11\pi}{3}\right)$ **5.** $\cot\left(-\frac{15\pi}{4}\right)$

Solution:

1. $\sin 765°$

We know that values of $\sin x$ repeat after an interval of 2π or $360°$

So we get

$\sin 765° = \sin(2 \times 360° + 45°)$

By further calculation

$= \sin 45° = 1/\sqrt{2}$

2. $\operatorname{cosec}(-1410°)$

We know that values of $\operatorname{cosec} x$ repeat after an interval of 2π or $360°$

So we get

$\operatorname{cosec}(-1410°) = \operatorname{cosec}(-1410° + 4 \times 360°)$

By further calculation

$= \operatorname{cosec}(-1410° + 1440°) = \operatorname{cosec} 30° = 2$

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3. $\tan\frac{19\pi}{3}$

We know that values of $tan\ x$ repeat after an interval of π or 180°

So we get

$\tan\frac{19\pi}{3} = \tan 6\frac{1}{3}\pi$

By further calculation , We get

$= \tan\left(6\pi + \frac{\pi}{3}\right) = \tan\frac{\pi}{3}$

$= \tan 60° = \sqrt{3}$

4. $\sin\left(-\frac{11\pi}{3}\right)$

We know that values of $sin\ x$ repeat after an interval of 2π or 360°

So we get

$\sin\left(-\frac{11\pi}{3}\right) = \sin\left(-\frac{11\pi}{3} + 2\times 2\pi\right)$

By further calculation

$= \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$

5. $\cot\left(-\frac{15\pi}{4}\right)$

We know that values of $\tan\ x$ repeat after an interval of π or 180°

So we get

$\cot\left(-\frac{15\pi}{4}\right) = \cot\left(-\frac{15\pi}{4} + 4\pi\right)$

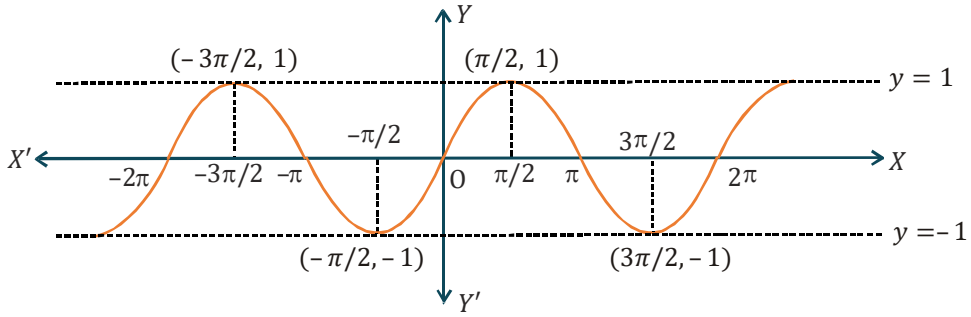
By further calculation

$= \cot\frac{\pi}{4} = 1$

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GRAPHS OF TRIGONOMETRIC RATIOS:

(1) $y = \sin x, x \in R, y \in [-1, 1]$



Periodic with period 2π

Note :

(i) $-1 \le \sin\theta \le 1$

(ii) $\sin 0 = 0;\ \sin\pi = 0;\ \sin 2\pi = 0$

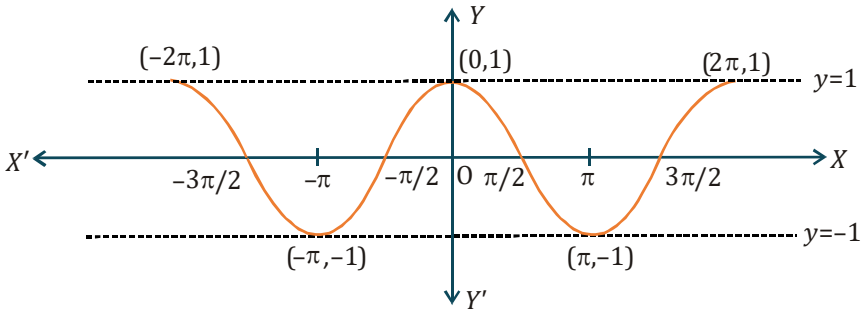
$\sin(n\pi) = 0, n \in I$

i.e. sine of integral multiple of $\pi = 0$

(iii) $\sin(2n\pi + \theta) = \sin\theta,\ n \in I$

For More Join: @IITJEE_Advanced

(2) $y = \cos x, x \in R, y \in [-1,1]$



Periodic with period 2π

Note:

(i) $-1 \le \cos\theta \le 1$

(ii) $\cos\frac{\pi}{2} = 0; \cos\frac{3\pi}{2} = 0; \cos\frac{5\pi}{2} = 0$

$\cos(2n-1)\frac{\pi}{2} = 0, n \in I$

i.e. cosine of odd integral multiple of $\frac{\pi}{2}$ is zero.

(iii) $\cos 0 = 1; \cos 2\pi = 1; \cos 4\pi = 1$

$\cos 2m\pi = 1, m \in I$

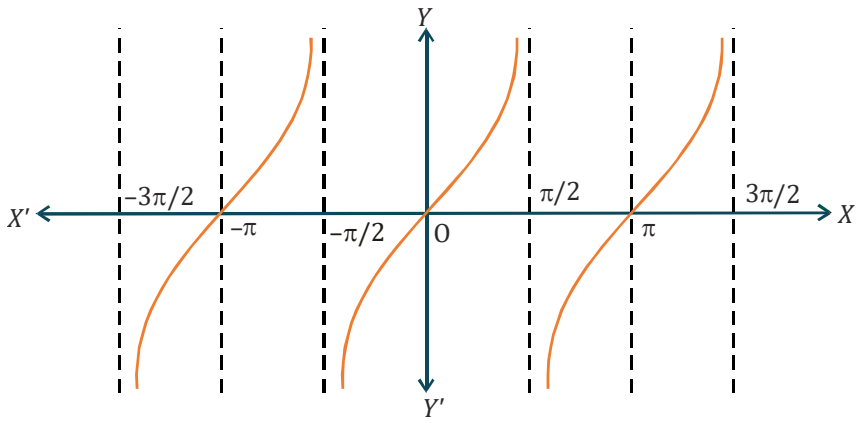
i.e. cos of even multiple of $\pi = 1$

(iv) $\cos\pi = -1; \cos 3\pi = -1; \cos 5\pi = -1$

$\cos(2m-1)\pi$ = cos of odd multiple of $\pi = -1$

$\cos(2n\pi + \theta) = \cos\theta, n \in I$

(3) $y = \tan x, x \ne \frac{\pi}{2}, \frac{3\pi}{2}$ ..... $\ne (2n-1)\frac{\pi}{2} : n \in I$



Periodic with period π

Note :

(i) $\tan(n\pi) = 0$

(ii) $\tan(n\pi + \theta) = \tan\theta, n \in I$

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DOMAINS, RANGES AND PERIODICITY OF TRIGONOMETRIC FUNCTIONS:

T-Ratio	Domain	Range	Period
$\sin x$	R	$[-1,1]$	2π
$\cos x$	R	$[-1,1]$	2π
$\tan x$	$R-\{(2n+1)\pi/2 ; n\in I\}$	R	π

Highlight of the Table:

(1) $-1 \le \sin\theta \le 1; -1 \le \cos\theta \le 1$

(2) $\sin\theta = 0, \sin\pi = 0, \sin 2\pi = 0$

$\Rightarrow \sin n\pi = 0$ where $n \in I$

$\therefore$ sine of integral multiple of π is zero

(3) $\tan 0 = 0, \tan\pi = 0$

$\Rightarrow \tan n\pi = 0$, where $n \in I$

$\therefore$ tangent of integral multiple of π is zero.

(4) $\cos\frac{\pi}{2} = 0;\ \cos\frac{3\pi}{2} = 0;\ \cos\left(-\frac{\pi}{2}\right) = 0$

$\Rightarrow \cos(2n-1)\frac{\pi}{2} = 0,$ where $n \in I$

cosine of odd integral multiple of $\frac{\pi}{2}$ is zero.

(5) $\cos 0 = 1, \cos\pi = -1, \cos 2\pi = 1$

$\therefore \cos 2m\pi = 1; \cos(2m-1)\pi = -1$, where $m \in I$

cosine of odd integral multiple of π is -1 and even integral multiple of π is 1. Same is the fate of sec.

(6) $\tan\frac{\pi}{2} = \text{ND};\ \tan\frac{3\pi}{2} = \text{ND}$

$\tan(2n+1)\frac{\pi}{2} = \text{ND},$ where $n \in I$

tangent of odd integral multiple of $\frac{\pi}{2}$ is ND

(7) $\sin\left(2n\pi + \frac{\pi}{2}\right) = 1, \sin\left(2n\pi - \frac{\pi}{2}\right) = -1 = \sin\left(2n\pi + \frac{3\pi}{2}\right)$

Illustration 17

Number of solution of the equation $(2\sin\theta - 1)(\sin\theta - 2) = 0$ in $(0, \pi)$

Solution:

$(2\sin\theta - 1)(\sin\theta - 2) = 0$ in $(0, \pi)$

$\Rightarrow \sin\theta = \frac{1}{2}$ OR $\sin\theta = 2$ (Not possible)

$\theta = 30°$ or $180° - 30° = 150°$

$\Rightarrow$ 2 solution

Illustration 18

Find values $\log(\cos(12346\pi)) = 0$

Solution:

$\log(\cos(12346\pi)) = \log(1) = 0$ [$\because \cos 2n\pi = 1$]

For More Join: @IITJEE_Advanced

Illustration 19

cos1 – sin1 > 0. True/False

[Ans. F]

Solution:

$1^c \approx 57.2°$

So if $45° < \theta < 90°$

$\sin\theta > \cos\theta$

So sin1 > cos1

Illustration 20

tan(11π) = 0 True/False

[Ans. T]

Solution:

$\tan(11\pi) = \tan(10\pi + \pi) = \tan\pi = 0$

$[\tan n\pi = 0]$

Illustration 21

$\frac{\sin 4}{\sin 6}$ is negative. True/False

[Ans. F]

Solution:

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$4^c \approx 4 \times 57° \approx 228°$

$6^c \approx 6 \times 57° \approx 342°$

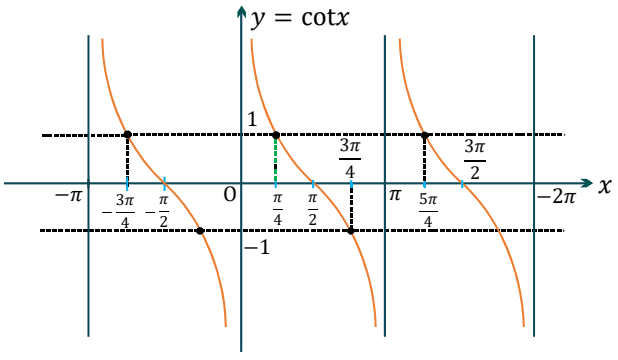
$\sin 228° = \sin(180° + 48°) = -\sin 48°$

$\sin 342° = \sin(360° - 18°) = -\sin 18°$

So $\frac{\sin 4}{\sin 6} = +ve$

GRAPHS OF TRIGONOMETRIC RATIOS:

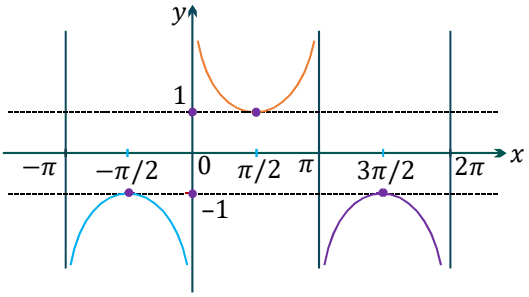
(1) $y = \cot x, x \in R, y \in (-\infty, \infty), x \neq n\pi$ for $n \in I$



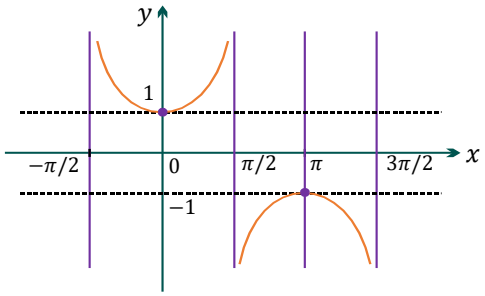
Note: $\cot(2n-1)\frac{\pi}{2} = 0 : n \in I$

For More Join: @IITJEE_Advanced

(2) $y = \text{cosec}\, x, y \in (-\infty, -1] \cup [1, \infty), x \in R - n\pi,\ x \neq n\pi$ for $n \in I$



(3) $y = \sec x, x \neq (2n+1)\frac{\pi}{2}$ for $n \in I$



DOMAINS, RANGES AND PERIODICITY OF TRIGONOMETRIC FUNCTIONS:

T-Ratio	Domain	Range	Period
$\cot x$	$R - \{n\pi : n \in I\}$	R	π
$\sec x$	$R - \{(2n+1)\pi/2 : n \in I\}$	$(-\infty,-1] \cup [1,\infty)$	2π
$\text{cosec}\, x$	$R - \{n\pi : n \in I\}$	$(-\infty,-1] \cup [1,\infty)$	2π

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Highlight of the Table:

(1) $\tan\theta, \cot\theta$ can attain all values from $(-\infty, \infty)$ i.e. any real value.

$\sec\theta$ & $\text{cosec}\,\theta \in (-\infty, -1]$ or $[1, \infty)$

(2) $\cot\frac{\pi}{2} = 0;\ \cot\frac{3\pi}{2} = 0$

$\therefore \cot(2n-1)\frac{\pi}{2} = 0,$ where $n \in I$

cotangent of odd integral multiple of $\frac{\pi}{2}$ is zero.

(3) $\cot 0° = \text{ND}, \cot\pi = \text{ND}, \cot 2\pi = \text{ND}$

$\therefore \cot n\pi = \text{ND}$, where $n \in I$

cotangent of integral multiple of π is not defined.

Illustration 22

If $\text{cosec}\theta = -2$ then find the two possible values of θ in $(0, 2\pi)$

Solution:

Given: $\text{cosec}\theta + 2 = 0$

$\Rightarrow \sin\theta = -\frac{1}{2}$

$\Rightarrow \theta = \left(\pi + \frac{\pi}{6}\right), \left(2\pi - \frac{\pi}{6}\right)$

$\Rightarrow \theta = 210°, 330°$

For More Join: @IITJEE_Advanced

Illustration 23

If secθ = 2 then solutions of equation in (0, 2π) is

Solution:

Given : $\sec\theta = 2$

$\Rightarrow cos\theta = \frac{1}{2}$

$\Rightarrow \theta = \left(\frac{\pi}{3}\right), \left(2\pi - \frac{\pi}{3}\right)$

$\Rightarrow \theta = 60°, 300°$

Illustration 24

If cotθ = 1 then solutions of equation in (0, 2π) is

Solution:

Given: $\cot\theta = 1$

$\Rightarrow \theta = \left(\frac{\pi}{4}\right), \left(\pi + \frac{\pi}{4}\right)$

$\Rightarrow \theta = 45°, 225°$

TRIGONOMETRIC RATIO OF COMPOUND ANGLES:

$A \pm B,\ A \pm B \pm C,\ \frac{A}{2} \pm \frac{B}{2}$ etc. are compound angles

$\sin(A + B) \neq \sin A + \sin B$

$\sin 90° \neq \sin 60° + \sin 30°$

$\sin(A + B) = \sin A\cos B + \cos A\sin B$

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Proof: From the figure

$\sin(A + B) = \frac{PM}{OP} = \frac{PR+RM}{OP}$

$= \frac{PR}{OP} + \frac{RM}{OP} = \frac{PR}{PN}\frac{PN}{OP} + \frac{NQ}{ON}\frac{ON}{OP}$

$= \cos A\sin B + \sin A\cos B$

$\Rightarrow \sin(A + B) = \sin A\cos B + \cos A\sin B$

Replace B by $-B$

$\sin(A + (-B)) = \sin A\cos(-B) + \cos A\sin(-B)$

$\therefore$ **$\sin(A - B) = \sin A\cos B - \cos A\sin B$**

Replace A by $\frac{\pi}{2} + A$

$\sin\left(\frac{\pi}{2} + A + B\right) = \sin\left(\frac{\pi}{2} + A\right)\cos B + \cos\left(\frac{\pi}{2} + A\right)\sin B$

or **$\cos(A + B) = \cos A\cos B - \sin A\sin B$**

$\cos(A - B) = \cos A\cos B + \sin A\sin B$

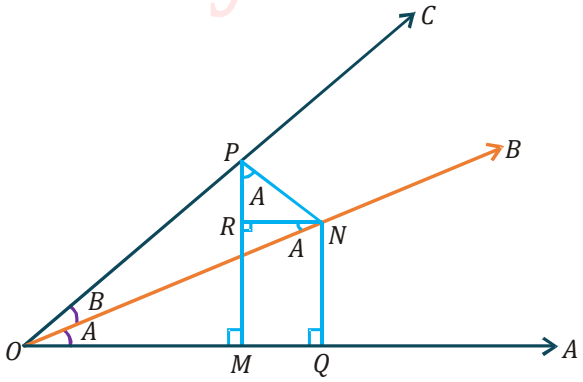


Illustration 25

$\sin 99°\cos 21° + \cos 99°\sin 21° =$

Solution:

$\sin(99° + 21°) = \sin 120° = \frac{\sqrt{3}}{2}$

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Illustration 26

The value of $\sin(n+1)A \sin(n+2)A + \cos(n+1)A \cos(n+2)A$ -

(A) $\sin(A+B)$ (B) $\cos(A+B)$ (C) $\cos A$ (D) $\sin A$

Ans. (C)

Solution:

We have $\sin(n+1)A \sin(n+2)A + \cos(n+1)A \cos(n+2)A$

$= \cos(n+1)A \cos(n+2)A + \sin(n+1)A \sin(n+2)A$

$= \cos[(n+1)A - (n+2)A] = \cos(-A) = \cos A$

VALUES OF TRIGONOMETRIC RATIOS OF SOME IMPORTANT ANGLES:

$\sin75° = \sin(45° + 30°) = \sin45°\cos30° + \cos45°\sin30° = \frac{1}{\sqrt{2}}.\frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}}.\frac{1}{2}$

or $\sin75° = \cos15° = \frac{1+\sqrt{3}}{2\sqrt{2}} = \frac{\sqrt{6}+\sqrt{2}}{4} = \frac{1}{\sqrt{6}-\sqrt{2}}$;

Similarly $\sin15° = \cos75° = \frac{\sqrt{3}-1}{2\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{4} = \frac{1}{\sqrt{6}+\sqrt{2}}$

Also $\tan75° = \frac{\sqrt{3}+1}{\sqrt{3}-1} = \frac{4+2\sqrt{3}}{2} = 2+\sqrt{3} = \cot15°$

$\tan15° = \frac{\sqrt{3}-1}{\sqrt{3}+1} = 2-\sqrt{3} = \cot75°$

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Note :

Remember value of sine & cosine of 15° & 75°

Illustration 27

$\cos\frac{2\pi}{3}\cos\frac{\pi}{4} - \sin\frac{2\pi}{3}\sin\frac{\pi}{4} = ?$

Solution:

$\cos\frac{11\pi}{12} = \cos\left(\pi - \frac{\pi}{12}\right) = -\cos\frac{\pi}{12} = -\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)$

Illustration 28

Prove that $\frac{\sin(x-y)}{\sin(x+y)} = \frac{\tan x - \tan y}{\tan x + \tan y}$

Solution:

$\frac{\sin(x-y)}{\sin(x+y)} = \frac{\sin x\cos y - \cos x\sin y}{\sin x\cos y + \cos x\sin y}$

$= \frac{\cos x\cos y(\tan x - \tan y)}{\cos x\cos y(\tan x + \tan y)} = \frac{\tan x - \tan y}{(\tan x + \tan y)}$

Illustration 29

If $\tan A = k \tan B$ prove that $\sin(A+B) = \frac{k+1}{k-1}\sin(A-B)$.

For More Join: @IITJEE_Advanced

Solution:

$\tan A = k\tan B \Rightarrow \frac{\tan A}{\tan B} = \frac{k}{1}$ using componendo & dividendo

$$\frac{\tan A + \tan B}{\tan A - \tan B} = \frac{k+1}{k-1} \Rightarrow \frac{\frac{\sin A}{\cos A} + \frac{\sin B}{\cos B}}{\frac{\sin A}{\cos A} - \frac{\cos B}{\cos B}} = \frac{k+1}{k-1}$$

$$\frac{\sin A\cos B + \cos A\sin B}{\sin A\cos B - \cos A\sin B} = \frac{k+1}{k-1}$$

$$\Rightarrow \sin(A+B) = \frac{(k+1)}{(k-1)}\sin(A-B)$$

Illustration 30

$\sin\alpha = \frac{3}{5}$ & $\cos\beta = \frac{9}{41}$. Find all possible values of $\sin(\alpha - \beta)$.

Solution:

Proper solution	Case	1	2	3	4	
	α	I	I	II	II	} Quadrant
	β	I	IV	I	IV	

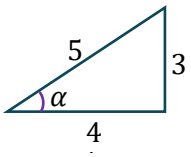
$\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$

Case (1) $= \frac{3}{5}\times\frac{9}{41} - \frac{4}{5}\times\frac{40}{41} = \frac{27-160}{205} = \frac{-133}{205}$

Case (2) $= \left(\frac{3}{5}\right)\times\frac{9}{41} - \left(\frac{4}{5}\right)\left(-\frac{40}{41}\right) = \frac{27+160}{205} = \frac{187}{205}$

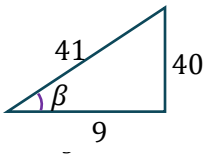
Case (3) $= \left(\frac{3}{5}\right)\times\left(\frac{9}{41}\right) - \left(-\frac{4}{5}\right)\left(\frac{40}{41}\right) = \frac{187}{205}$

Case (4) $= \left(\frac{3}{5}\right)\times\left(\frac{9}{41}\right) - \left(-\frac{4}{5}\right)\left(-\frac{40}{41}\right) = \frac{-133}{205}$



$\sin\alpha = \frac{3}{5}$

$\cos\alpha = \frac{4}{5}$



$\sin\beta = \frac{40}{41}$

$\cos\beta = \frac{9}{41}$

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FORMULAE & IDENTITIES:

(a) $\sin(A+B).\sin(A-B) = \sin^2A - \sin^2B = \cos^2B - \cos^2A$

Proof : $(\sin A\cos B + \cos A\sin B)(\sin A\cos B - \cos A\sin B)$

$= \sin^2A\cos^2B - \cos^2A\sin^2B = \sin^2A(1-\sin^2B) - (1-\sin^2A)\sin^2B$

$= \sin^2A - \sin^2B = (1-\cos^2A) - (1-\cos^2B) = \cos^2B - \cos^2A$

Illustration 31

Prove that $\sin^2\left(\frac{\pi}{8}+\frac{A}{2}\right) - \sin^2\left(\frac{\pi}{8}-\frac{A}{2}\right) = \frac{1}{\sqrt{2}}\sin A$

Solution:

$$\sin^2\left(\frac{\pi}{8}+\frac{A}{2}\right) - \sin^2\left(\frac{\pi}{8}-\frac{A}{2}\right) = \sin\left(\frac{\pi}{8}+\frac{A}{2}+\frac{\pi}{8}-\frac{A}{2}\right).\sin\left(\frac{\pi}{8}+\frac{A}{2}-\frac{\pi}{8}+\frac{A}{2}\right)$$

$$= \sin\frac{\pi}{4}.\sin A = \frac{1}{\sqrt{2}}\sin A$$

(b) $\cos^2A - \sin^2B = \cos(A+B)\cos(A-B) = \cos^2B - \sin^2A$

Illustration 32

$\cos^2\left(\frac{\pi}{4}+\theta\right) - \sin^2\left(\frac{\pi}{4}-\theta\right) = 0$

Solution:

$$\cos\left(\frac{\pi}{4}+\theta+\frac{\pi}{4}-\theta\right)\cos\left(\frac{\pi}{4}+\theta-\frac{\pi}{4}+\theta\right) = \cos\frac{\pi}{2}\cdot\cos 2\theta = 0$$

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FORMULA TO TRANSFORM THE PRODUCT INTO SUM OR DIFFERENCE:

(1) $2\sin A\cos B = \sin(A+B) + \sin(A-B)$

(2) $2\cos A\sin B = \sin(A+B) - \sin(A-B)$

(3) $2\cos A\cos B = \cos(A+B) + \cos(A-B)$

(4) $2\sin A\sin B = \cos(A-B) - \cos(A+B)$

Proof :

(1) $2\sin A\cos B = \sin(A+B) + \sin(A-B)$

$= \sin A\cos B + \cos A\sin B + \sin A\cos B - \cos A\sin B = 2\sin A\cos B$

Very Frequently used Formulae:

(i) $2\sin3\theta\cos\theta = \sin(3\theta+\theta) + \sin(3\theta-\theta) = \sin4\theta + \sin2\theta$

(ii) $2\cos11\theta\cos\theta = \cos(11\theta+\theta) + \cos(11\theta-\theta) = \cos12\theta + \cos10\theta$

(iii) $2\sin5\theta\sin3\theta = \cos(5\theta-3\theta) - \cos(5\theta+3\theta) = \cos2\theta - \cos8\theta$

Illustration 33

If $\theta = 7.5°$. Find $\dfrac{\sin8\theta\cos\theta - \sin6\theta\cos3\theta}{\cos2\theta\cos\theta - \sin3\theta\sin4\theta}$

Solution:

Multiply & divide by 2

$$\Rightarrow \frac{2\sin8\theta\cos\theta - 2\sin6\theta\cos3\theta}{2\cos2\theta\cos\theta - 2\sin3\theta\sin4\theta}$$

$$= \frac{\sin9\theta + \sin7\theta - (\sin9\theta + \sin3\theta)}{\cos3\theta + \cos\theta - (\cos\theta - \cos7\theta)} = \frac{\sin7\theta - \sin3\theta}{\cos3\theta + \cos7\theta}$$

$$= \frac{2\cos5\theta\sin2\theta}{2\cos5\theta\cos2\theta} = \tan2\theta = \tan15° = 2 - \sqrt{3}$$

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Illustration 34

Prove that $\cos^2 73 + \cos^2 47 + \cos73\cos47 = \dfrac{3}{4}$

Solution:

$$\cos^2 73° + 1 - \sin^2 47° + \frac{2\cos73°\cos47°}{2}$$

$$= (1 + \cos120°\cos26°) + \frac{\cos120° + \cos26°}{2}$$

$$= 1 - \frac{\cos26}{2} + \frac{\cos26}{2} - \frac{1}{4} = \boxed{\frac{3}{4}}$$

Illustration 35

$2\sin\left(\dfrac{5\pi}{12}\right)\sin\left(\dfrac{\pi}{12}\right) =$

(A) $-\dfrac{1}{2}$ (B) $\dfrac{1}{2}$ (C) $\dfrac{1}{4}$ (D) $\dfrac{1}{6}$

Ans. (B)

Solution:

$$\cos\left(\frac{5\pi}{12} - \frac{\pi}{12}\right) - \cos\left(\frac{5\pi}{12} + \frac{\pi}{12}\right)$$

$$= \cos\frac{\pi}{3} - \cos\frac{\pi}{2} = \frac{1}{2}$$

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Illustration 36

Prove that $\sin(45 + A)\sin(45 - A) = \frac{1}{2}\cos 2A$

Solution:

$= \frac{1}{2}[2\sin(45 + A)\sin(45 - A)]$

$= \frac{1}{2}[\cos(45 + A - 45 + A) - \cos(90)]$

$= \frac{1}{2}[\cos 2A]$

Illustration 37

$2\cos\frac{\pi}{13}.\cos\frac{9\pi}{13} + \cos\frac{3\pi}{13} + \cos\frac{5\pi}{13} = 0$

Solution:

$\cos\frac{10\pi}{13} + \cos\frac{8\pi}{13} + \cos\frac{3\pi}{13} + \cos\frac{5\pi}{13} = 0$

IDENTITIES FOR CONVERTING SUM TO PRODUCT:

Let $A + B = C$ & $A - B = D$

$\therefore A = \frac{C + D}{2}, B = \frac{C - D}{2}$

1. $\sin C + \sin D = 2\sin\left(\frac{C + D}{2}\right)\cos\left(\frac{C - D}{2}\right)$
2. $\sin C - \sin D = 2\cos\left(\frac{C + D}{2}\right)\sin\left(\frac{C - D}{2}\right)$
3. $\cos C + \cos D = 2\cos\left(\frac{C + D}{2}\right)\cos\left(\frac{C - D}{2}\right)$
4. $\cos C - \cos D = -2\sin\left(\frac{C + D}{2}\right)\sin\left(\frac{C - D}{2}\right) = 2\sin\left(\frac{C + D}{2}\right)\sin\left(\frac{D - C}{2}\right)$

Proof:

1. $2\sin A\cos B = \sin(A + B) + \sin(A - B)$

$= 2\sin\left(\frac{C + D}{2}\right)\cos\left(\frac{C - D}{2}\right)$

$= \sin\left(\frac{C + D}{2} + \frac{C - D}{2}\right) + \sin\left(\frac{C + D}{2} - \frac{C - D}{2}\right) = \sin C + \sin D$

Illustration 38

(i) $\frac{\sin 7\theta - \sin 5\theta}{\cos 7\theta + \cos 5\theta} = \tan\theta$ (ii) $\frac{\sin A + \sin 3A}{\cos A - \cos 3A} = \cot A$

Solution:

(i) $\text{LHS} = \frac{2\cos 6\theta \cdot \sin\theta}{2\cos 6\theta \cdot \cos\theta} = \tan\theta$

(ii) $\frac{2\sin 2A\cos A}{2\sin 2A\sin A} = \cot A$

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Illustration 39

$\frac{\sin\theta + \sin3\theta + \sin5\theta}{\cos\theta + \cos3\theta + \cos5\theta} =$

(A) tanθ (B) tan 5θ (C) tan 3θ (D) None of these

Ans. (C)

Solution:

$$\frac{(\sin\theta + \sin5\theta) + \sin3\theta}{(\cos\theta + \cos5\theta) + \cos3\theta} = \frac{2\sin\frac{\theta+5\theta}{2}.\cos\frac{\theta-5\theta}{2} + \sin3\theta}{2\cos\frac{\theta+5\theta}{2}.\cos\frac{\theta-5\theta}{2} + \cos3\theta}$$

$$= \frac{2\sin3\theta\cos(-2\theta) + \sin3\theta}{2\cos3\theta\cos(-2\theta) + \cos3\theta} = \frac{\sin3\theta[2\cos2\theta + 1]}{\cos3\theta[2\cos\theta + 1]} = \frac{\sin3\theta}{\cos3\theta} \Rightarrow \tan3\theta$$

Illustration 40

$\frac{\sin A+\sin3A+\sin5A+\sin7A}{\cos A+\cos3A+\cos5A+\cos7A}$ is equal to -

(A) sin $4A$ (B) cos $4A$ (C) tan $4A$ (D) None of these

Ans. (C)

Solution:

$$\frac{(\sin A + \sin7A) + (\sin3A + \sin5A)}{(\cos A + \cos7A) + (\cos3A + \cos5A)} = \frac{2\sin4A\cos3A + 2\sin4A\cos A}{2\cos4A\cos3A + 2\cos4A\cos A}$$

$$= \frac{2\sin4A(\cos3A + \cos A)}{2\cos4A(\cos3A + \cos A)} = \tan4A$$

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Illustration 41

Prove that sin50° – sin70° + sin10° = 0

Solution:

$2\cos60°\sin(-10°) + \sin10° = 2 \times \frac{1}{2}(-\sin10°) + \sin10° = 0.$

Illustration 42

If three angles A,B,C are in A.P. then the value of $\frac{\sin A-\sin C}{\cos C-\cos A}$, (if defined) is equal to -

(A) $\cot\left(\frac{A-C}{2}\right)$ (B) cot B (C) tan B (D) None of these

Ans. (B)

Solution:

Angles A, B, C are in A.P. then $2B = A + C$

Now $\frac{\sin A - \sin C}{\cos C - \cos A} = \frac{2\cos\frac{A+C}{2}\sin\frac{A-C}{2}}{2\sin\frac{A+C}{2}\sin\frac{A-C}{2}} = \cot\frac{A+C}{2} = \cot\frac{2B}{2} = cot\ B$

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Illustration 43

Prove that $\cos\alpha + \cos\beta + \cos\gamma + \cos(\alpha+\beta+\gamma) = 4\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\beta+\gamma}{2}\right)\cos\left(\frac{\gamma+\alpha}{2}\right)$

Solution:

$\underbrace{\cos\alpha + \cos\beta} + \underbrace{\cos\gamma + \cos}(\alpha+\beta+\gamma) = 4\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\beta+\gamma}{2}\right)\cos\left(\frac{\gamma+\alpha}{2}\right)$

L.H.S.$=2\cos\left(\frac{\alpha+\beta}{2}\right)\cdot\cos\left(\frac{\alpha-\beta}{2}\right)+2\cos\left(\frac{\alpha+\beta+2\gamma}{2}\right)\cdot\cos\left(\frac{\alpha+\beta}{2}\right)$

$= 2\cos\left(\frac{\alpha+\beta}{2}\right)\left[\cos\left(\frac{\alpha-\beta}{2}\right)+\cos\left(\frac{\alpha+\beta+2\gamma}{2}\right)\right]$

$= 2\cos\left(\frac{\alpha+\beta}{2}\right)2\cos\left(\frac{\alpha-\beta+\alpha+\beta+2\gamma}{4}\right)\cos\left(\frac{\alpha+\beta+2\gamma-\alpha+\beta}{4}\right)$

$= 4\cos\left(\frac{\alpha+\beta}{2}\right)\cdot\cos\left(\frac{\beta+\gamma}{2}\right)\cdot\cos\left(\frac{\gamma+\alpha}{2}\right)$

Illustration 44

Prove that $\frac{\cos20^\circ+\sin10^\circ}{\cos20^\circ-\sin10^\circ} = \sqrt{3}\tan40^\circ$

Solution:

$\frac{\sin70^\circ+\sin10^\circ}{\sin70^\circ-\sin10^\circ}$

$= \frac{2\sin\frac{(70^\circ+10^\circ)}{2}.\cos\frac{(70^\circ-10^\circ)}{2}}{2\cos\frac{(70^\circ+10^\circ)}{2}.\sin\frac{(70^\circ-10^\circ)}{2}}$

$= \frac{2\sin40^\circ\cos30^\circ}{2\cos40^\circ\sin30^\circ} = \tan40^\circ\times\frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}\tan40^\circ$

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$$\tan(A+B) = \frac{\tan A+\tan B}{1-\tan A\tan B}$$

Proof : $\tan(A+B) = \frac{\sin(A+B)}{\cos(A+B)} = \frac{\sin A\cos B+\cos A\sin B}{\cos A\cos B-\sin A\sin B}$ Divide by $\cos A\cos B$

$\Rightarrow \tan(A+B) = \frac{\tan A+\tan B}{1-\tan A\tan B}$

Similarly $\tan(A-B) = \frac{\tan A-\tan B}{1+\tan A\tan B}$

Also, $\cot(A+B) = \frac{\cot A\cot B-1}{\cot B+\cot A}, \cot(A-B) = \frac{\cot A\cot B+1}{\cot B-\cot A}$

Note : (i) $\tan\left(\frac{\pi}{4}+A\right) = \frac{1+\tan A}{1-\tan A}$ (ii) $\tan\left(\frac{\pi}{4}-A\right) = \frac{1-\tan A}{1+\tan A}$

Illustration 45

If $\tan A = -\frac{1}{2}$ and $\tan B = -\frac{1}{3}$ then $A+B$ can be

(A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) None

Solution:

Using the $\tan(A+B) = \frac{\tan A+\tan B}{1-\tan A\tan B} = \frac{-\frac{1}{2}-\frac{1}{3}}{1-\frac{1}{2}\cdot\frac{1}{3}} = \frac{-\frac{5}{6}}{\frac{5}{6}} = -1 \Rightarrow A+B = \frac{3\pi}{4}$

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Illustration 46

Prove that $\frac{\cos9°+\sin9°}{\cos9°-\sin9°} = \tan54°$

Divide by $\cos9° = \frac{1+\tan9°}{1-\tan9°} = \tan(45°+9°)$

Illustration 47

Prove that $\cot16°\cot44° + \cot44°\cot76° - \cot76°\cot16° = 3$

Solution:

$(\cot16°\cot44° - 1) + (\cot44°\cot76° - 1) - (\cot76°\cot16° + 1)$

$= \cot60°(\cot44° + \cot16°) + \cot120°(\cot44° + \cot76°) - \cot60°(\cot16° - \cot76°)$

$= \frac{1}{\sqrt{3}}[\cot44° + \cot16° - \cot44° - \cot76° - \cot16° + \cot76°] = 0$

Illustration 48

Prove that $\tan70° = \cot70° + 2\cot40°$.

Solution:

L.H.S. $= \tan70° = \tan(20°+50°) = \frac{\tan20°+\tan50°}{1-\tan20°\tan50°}$

or $\tan70° - \tan20°\tan50°\tan70° = \tan20° + \tan50°$

or $\tan70° = \tan70°\tan50°\tan20° + \tan20° + \tan50° = 2\tan50° + \tan20°$

$= \cot70° + 2\cot40° =$ R.H.S.

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Illustration 49

$\tan3A.\tan2A\tan A = \tan3A - \tan2A - \tan A$

Solution:

$\because \tan3A = \tan(A+2A) = \frac{\tan A + \tan2A}{1-\tan A\tan2A}$

$\Rightarrow \tan3A - \tan A\tan2A\tan3A = \tan A + \tan2A$

$\Rightarrow \tan3A - \tan2A - \tan A = \tan A\tan2A\tan3A$

Illustration 50

$\cot x\cot2x - \cot2x\cot3x - \cot3x\cot x = 1$

Solution:

$\cot x\cot2x\cot3x[\tan3x - \tan x - \tan2x]$

$\cot x\cot2x\cot3x[\tan x\tan2x\tan3x] = 1$

Illustration 51

If $A + B = 45°$. Find value of $(1+\tan A)(1+\tan B)$

Solution:

$(1+\tan A)(1+\tan B) = 1 + \tan A + \tan B + \tan A\tan B$

$= 1 + \tan(A+B)(1-\tan A\tan B) + \tan A\tan B$ $\quad \begin{cases} \because & \tan(A+B) = \frac{\tan A+\tan B}{1-\tan A\tan B} \\ \Rightarrow & \tan A + \tan B = 1 - \tan A\tan B \end{cases}$

$= 1 + 1 - \tan A\tan B + \tan A\tan B = 2$

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TRIGONOMETRICAL RATIOS OF THE SUM OF MORE THAN 2 ANGLES:

$\sin(A+B+C) = \sin(A+B)\cos C + \cos(A+B)\sin C$

$= [\sin A\cos B + \cos A\sin B]\cos C + [\cos A\cos B - \sin A\sin B]\sin C$

$= \sin A\cos B\cos C + \cos A\sin B\cos C + \cos A\cos B\sin C - \sin A\sin B\sin C$

$\cos(A+B+C) = \cos(A+B)\cos C - \sin(A+B)\sin C$

$= (\cos A\cos B - \sin A\sin B)\cos C - (\sin A\cos B + \cos A\sin B)\sin C$

$= \cos A\cos B\cos C - \cos A\sin B\sin C - \sin A\cos B\sin C - \sin A\sin B\cos C.$

$$\text{Also, } \tan(A+B+C) = \frac{\tan(A+B)+\tan C}{1-\tan(A+B)\tan C} = \frac{\frac{\tan A+\tan B}{1-\tan A\tan B}+\tan C}{1-\frac{\tan A+\tan B}{1-\tan A\tan B}\tan C}$$

$$= \frac{\tan A+\tan B+\tan C-\tan A\tan B\tan C}{1-\tan B\tan C-\tan C\tan A-\tan A\tan B} = \frac{s_1-s_3}{1-s_2}$$

$$\tan(A_1+A_2+\ldots+A_n) = \frac{s_1-s_3+s_5-s_7+\ldots.}{1-s_2+s_4-s_6+\ldots.}$$

where $s_1 = \tan A_1 + \tan A_2 + \ldots.. + \tan A_n$ = the sum of the tangent of the separate angles,

$s_2 = \tan A_1\tan A_2 + \tan A_1\tan A_3 + \ldots..$ = the sum of the tangents taken two at a time,

$s_3 = \tan A_1\tan A_2\tan A_3 + \tan A_2\tan A_3\tan A_4 + \ldots..$

= the sum of the tangents taken three at a time, and so on.

TRIGONOMETRIC RATIOS OF MULTIPLE ANGLES:

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Trigonometrical Ratios of an Angle 2θ in Terms of the Angle:

(i) $\sin 2\theta = 2\sin\theta\cos\theta = \frac{2\tan\theta}{1+\tan^2\theta}$

(ii) $\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta}$

(iii) $1 + \cos 2\theta = 2\cos^2\theta$

(iv) $1 - \cos 2\theta = 2\sin^2\theta$

(v) $\tan\theta = \frac{1-\cos 2\theta}{\sin 2\theta} = \frac{\sin 2\theta}{1+\cos 2\theta}$

(vi) $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$

1. $\mathbf{\sin 2A = 2\sin A\cos A = \frac{2\tan A}{1+\tan^2 A}}$

Proof :

$$\sin 2A = \sin(A+A) = \sin A\cos A + \sin A\cos A = 2\sin A\cos A = \frac{2\sin A\cos A}{\sin^2 A+\cos^2 A} = \frac{2\tan A}{1+\tan^2 A}$$

Ex. $\sin 4A = 2\sin 2A\cos 2A$

$\sin 100A = 2\sin 50A\cos 50A$

$\sin A = 2\sin\frac{A}{2}\cos\frac{A}{2}$

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2. $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A = \dfrac{1 - \tan^2 A}{1 + \tan^2 A}$

Proof :

$\cos(A + A) = \cos A \cos A - \sin A \cdot \sin A = \cos^2 A - \sin^2 A = \dfrac{\cos^2 A - \sin^2 A}{\cos^2 A + \sin^2 A}$

Also $\cos 2A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$

Illustration 52

If $\sin A = \frac{4}{5}$. Find $\cos 2A$

Solution:

$\cos 2A = 1 - 2\sin^2 A = 1 - 2 \times \frac{16}{25} = -\frac{7}{25}$

Example:

(i) $\dfrac{1+\cos 2\theta}{\sin 2\theta} = \dfrac{2\cos^2\theta}{2\sin\theta\cos\theta} = \cot\theta$ **(ii)** $\dfrac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$ **(iii)** $\dfrac{1-\cos 2\theta}{1+\cos 2\theta} = \dfrac{2\sin^2\theta}{2\cos^2\theta} = \tan^2\theta$

Illustration 53

Prove that $\dfrac{\sec 8A - 1}{\sec 4A - 1} = \dfrac{\tan 8A}{\tan 2A}$

Solution:

$= \dfrac{1 - \cos 8A}{1 - \cos 4A} \times \dfrac{\cos 4A}{\cos 8A} = \dfrac{2\sin^2 4A}{2\sin^2 2A} \cdot \dfrac{\cos 4A}{\cos 8A} = \dfrac{\sin 8A}{\cos 8A} \cdot \dfrac{\sin 4A}{2\sin^2 2A}$

$= \tan 8A \cdot \dfrac{2\sin 2A \cos 2A}{2\sin^2 2A} = \dfrac{\tan 8A}{\tan 2A}$

3. $\sin 3A = 3\sin A - 4\sin^3 A$

Proof :

$\sin 3A = \sin(A + 2A) = \sin A \cos 2A + \cos A \sin 2A = \sin A (1 - 2\sin^2 A) + 2\sin A (1 - \sin^2 A)$

$= 3\sin A - 4\sin^3 A$

Similarly, $\cos 3A = 4\cos^3 A - 3\cos A$

& $\tan 3A = \dfrac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$

Illustration 54

$\sin 90° = 3\sin 30° - 4\sin^3 30°$

Solution:

$1 = 3 \cdot \frac{1}{2} - 4 \cdot \frac{1}{8} = 1$

Illustration 55

$\dfrac{3\cos\theta + \cos 3\theta}{3\sin\theta - \sin 3\theta}$ is equal to-

(A) $1 + \cot^2\theta$ (B) $\cot^4\theta$ (C) $\cot^3\theta$ (D) $2\cot\theta$

Ans. (C)

Solution:

$\dfrac{3\cos\theta + 4\cos^3\theta - 3\cos\theta}{3\sin\theta - 3\sin\theta + 4\sin^3\theta} = \cot^3\theta$

Illustration 56

Find the value of $-6 \sin 40° + 8 \sin^3 40°$

Solution:

$-2(3\sin 40° - 4\sin^3 40°) = -2 \sin 120° = -\sqrt{3}$

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Illustration 57

Prove that $(4\cos^2 9° - 3)(4\cos^2 27° - 3) = \tan 9°$

Solution:

$$\frac{(4\cos^3 9° - 3\cos 9°)}{\cos 9°}\frac{(4\cos^3 27° - 3\cos 27°)}{\cos 27°} = \frac{\cos 27°}{\cos 9°} \times \frac{\cos 81°}{\cos 27°} = \frac{\sin 9°}{\cos 9°} = \tan 9°$$

TRIGONOMETRIC RATIOS OF SUB MULTIPLE ANGLES:

Since the trigonometric relations are true for all values of angle θ, they will be true if instead of θ substitute $\frac{\theta}{2}$

(i) $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} = \frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$

(ii) $\cos\theta = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} = 2\cos^2\frac{\theta}{2} - 1 = 1 - 2\sin^2\frac{\theta}{2} = \frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$

(iii) $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$

(iv) $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$

(v) $\tan\frac{\theta}{2} = \frac{1-\cos\theta}{\sin\theta} = \frac{\sin\theta}{1+\cos\theta}$

(vi) $\tan\theta = \frac{2\tan\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}}$

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(vii) $\sin\frac{\theta}{2} = \pm\sqrt{\frac{1-\cos\theta}{2}}$

(viii) $\cos\frac{\theta}{2} = \pm\sqrt{\frac{1+\cos\theta}{2}}$

(ix) $\tan\frac{\theta}{2} = \pm\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$

for (vii) to (ix), we decide the sign of ratio according to value of θ.

Illustration 58

$\sin 67\frac{1}{2}^\circ + \cos 67\frac{1}{2}^\circ$ is equal to

(A) $\frac{1}{2}\sqrt{4+2\sqrt{2}}$ (B) $\frac{1}{2}\sqrt{4-2\sqrt{2}}$ (C) $\frac{1}{4}\left(\sqrt{4+2\sqrt{2}}\right)$ (D) $\frac{1}{4}\left(\sqrt{4-2\sqrt{2}}\right)$

Ans. (A)

Solution:

$$\sin 67\frac{1}{2}^\circ + \cos 67\frac{1}{2}^\circ = \sqrt{1+\sin 135°} = \sqrt{1+\frac{1}{\sqrt{2}}} \quad (\text{using } \cos A + \sin A = \sqrt{1+\sin 2A})$$

$$= \frac{1}{2}\sqrt{4+2\sqrt{2}}$$

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Illustration 59

Prove that $\cos^4\frac{A}{2} - \sin^4\frac{A}{2} = \cos A$

Solution:

$= \left(\cos^2\frac{A}{2} - \sin^2\frac{A}{2}\right)\left(\cos^2\frac{A}{2} + \sin^2\frac{A}{2}\right) = cosA$

Illustration 60

Value of $\sqrt{2+\sqrt{2+2\cos 4\theta}} =$

(A) $2\sin\theta$ (B) $2\cos\theta$ (C) $\sin\theta$ (D) $\cos\theta$

Ans. (B)

Solution:

$$\sqrt{2+\sqrt{2+2\cos 4\theta}} = \sqrt{2+\sqrt{2(1+\cos 4\theta)}} = \sqrt{2+\sqrt{2.2\cos^2 2\theta}}$$

$$= \sqrt{2+2\cos 2\theta} = \sqrt{2(1+\cos 2\theta)} = \sqrt{2.2\cos^2\theta} = 2\cos\theta$$

Illustration 61

Evaluate: (i) $\sin 22.5°$ (ii) $\cos 22.5°$ (iii) $\tan 22.5°$

Solution:

(i) $\sin^2(22.5°) = \frac{1-\cos 45°}{2} = \frac{1-\frac{1}{\sqrt{2}}}{2} = \frac{\sqrt{2}-1}{2\sqrt{2}}$

$= \frac{2-\sqrt{2}}{4} \Rightarrow \sin 22.5° = \frac{\sqrt{2-\sqrt{2}}}{2} = \cos 67\frac{1}{2}$

(ii) Similarly, $\cos(22.5°) = \frac{\sqrt{2+\sqrt{2}}}{2} = \sin 67\frac{1}{2}^\circ$

(iii) $\tan 22\frac{1}{2}^o = \sqrt{\frac{1-\cos 45°}{1+\cos 45°}} = \frac{\sqrt{1-\frac{1}{\sqrt{2}}}}{\sqrt{1+\frac{1}{\sqrt{2}}}} = \sqrt{\frac{\sqrt{2}-1}{\sqrt{2}+1}} \times \sqrt{\frac{\sqrt{2}-1}{\sqrt{2}-1}} = \sqrt{2}-1 = \tan\frac{\pi}{8}$ (learn)

Also $\sqrt{2}+1 = \cot\frac{\pi}{8}$ (learn)

Illustration 62

Prove $\tan 7\frac{1}{2}^o = \sqrt{6}-\sqrt{4}-\sqrt{3}+\sqrt{2}$

Solution:

$$= \frac{2\sin 7\frac{1}{2}}{2\cos 7\frac{1}{2}} \cdot \frac{\sin 7\frac{1}{2}}{\sin 7\frac{1}{2}} = \frac{1-\cos 15}{\sin 15} = \frac{1-\frac{\sqrt{3}+1}{2\sqrt{2}}}{\frac{\sqrt{3}-1}{2\sqrt{2}}}$$

$$= \frac{2\sqrt{2}-(\sqrt{3}+1)}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \sqrt{6}-\sqrt{4}-\sqrt{3}+\sqrt{2}$$

Similarly $\cot 7\frac{1}{2}^\circ = \sqrt{2}+\sqrt{3}+\sqrt{4}+\sqrt{6}$

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SOME IMPORTANT RESULTS:

(i) $\sin\theta\sin(60°-\theta)\sin(60°+\theta) = \frac{1}{4}\sin3\theta$

Proof:

$\frac{1}{2}\sin\theta[2\sin(60°-\theta)\sin(60°+\theta)]$

$=\frac{1}{2}\sin\theta[\cos2\theta-\cos120°]$

$=\frac{1}{2}\sin\theta\left[\cos2\theta+\frac{1}{2}\right]$

$=\frac{1}{4}[2\sin\theta\cos2\theta+\sin\theta]$

$=\frac{1}{4}[\sin3\theta-\sin\theta+\sin\theta]$

$=\frac{1}{4}\sin3\theta$

(ii) $\cos\theta\cos(60°-\theta)\cos(60°+\theta) = \frac{1}{4}\cos3\theta$

(iii) $\tan\theta\tan(60°-\theta)\tan(60°+\theta) = \tan3\theta$

Illustration 63

$\sin20°\sin40°\sin60°\sin80° = \frac{3}{16}$

Solution:

Hint : $(\sin20°\sin40°\sin80°)\sin60° = \frac{1}{4}\sin60°\cdot\sin60° = \frac{3}{16}$

T-RATIO FOR SOME STANDARD ANGLES:

18°, 36°, 54°, 72°

$$\boxed{\sin18° = \sin\frac{\pi}{10} = \frac{\sqrt{5}-1}{4} = \cos72° = \cos\frac{2\pi}{5}}$$

Proof: Let $\theta = 18° \Rightarrow 5\theta = 90° \Rightarrow 2\theta = 90° - 3\theta$

Now $\sin(2\theta) = \sin(90° - 3\theta) = \cos3\theta$

$\Rightarrow 2\sin\theta\cos\theta = 4\cos^3\theta - 3\cos\theta \Rightarrow 2\sin\theta = 4\cos^2\theta - 3$

$\Rightarrow 2\sin\theta = 1 - 4\sin^2\theta \Rightarrow 4\sin^2\theta + 2\sin\theta - 1 = 0$

$\Rightarrow \sin\theta = \frac{-2\pm2\sqrt{5}}{8} \quad \therefore \sin18° = \frac{\sqrt{5}-1}{4} \; ; \; \frac{-\sqrt{5}-1}{4}$ is not possible as $\sin18° > 0$.

Now $\cos36° = 1 - 2\sin^2 18° = 1 - 2\left(\frac{\sqrt{5}-1}{4}\right)^2$

or $\cos36° = \cos\pi/5 = \frac{\sqrt{5}+1}{4} = \sin54° = \sin3\pi/10$.

Now $\cos18° = \sqrt{1-\sin^2 18°} = \sqrt{1-\left(\frac{\sqrt{5}-1}{4}\right)^2}$

or $\cos18° = \frac{\sqrt{10+2\sqrt{5}}}{4} = \cos\frac{\pi}{10} = \sin72 = \sin^2\frac{\pi}{5}$

$\sin36° = \sqrt{1-\cos^2 36°} = \sqrt{1-\left(\frac{\sqrt{5}+1}{4}\right)^2}$

$\Rightarrow \sin36° = \frac{\sqrt{10-2\sqrt{5}}}{4} = \cos54°$

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Illustration 64

Find the value of $\cos^2 48° - \sin^2 12°$

Solution:

$$\cos(60°)\cos(36°) = \frac{\sqrt{5}+1}{4} \times \frac{1}{2} = \frac{\sqrt{5}+1}{8}$$

Illustration 65

Find the value of $\sin(132°)\sin(12°)$

Solution:

$$\frac{1}{2}[\cos(120°) - \cos(144°)] = \frac{1}{2}\left[-\frac{1}{2} + \cos 36°\right] = \frac{1}{2}\left[\frac{\sqrt{5}+1}{4} - \frac{1}{2}\right] = \frac{\sqrt{5}-1}{8}$$

TRIGONOMETRIC RATIOS FOR SOME STANDARD ANGLES:

(i) $\sin 15° = \sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}} = \cos 75° = \cos\frac{5\pi}{12}$

(ii) $\cos 15° = \cos\frac{\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}} = \sin 75° = \sin\frac{5\pi}{12}$

(iii) $\tan 15° = \tan\frac{\pi}{12} = 2 - \sqrt{3} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \cot 75° = \cot\frac{5\pi}{12}$

(iv) $\tan 75° = \tan\frac{5\pi}{12} = 2 + \sqrt{3} = \frac{\sqrt{3}+1}{\sqrt{3}-1} = \cot 15° = \cot\frac{\pi}{12}$

(v) $\tan(22.5°) = \tan\frac{\pi}{8} = \sqrt{2} - 1 = \cot(67.5°) = \cot\frac{3\pi}{8}$

(vi) $\tan(67.5°) = \tan\frac{3\pi}{8} = \sqrt{2} + 1 = \cot(22.5°) = \cot\frac{\pi}{8}$

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Illustration 66

$\sin^2 72° - \sin^2 60° = \frac{\sqrt{5}-1}{8}$

Solution:

$$1 - \cos^2 72° - \frac{3}{4} = \frac{1}{4} - \left(\frac{\sqrt{5}-1}{4}\right)^2 = \frac{\sqrt{5}-1}{8}$$

TRIGONOMETRIC IDENTITIES IN A TRIANGLE (CONDITIONAL IDENTITIES):

If A, B, C are the angles of a triangle i.e. $A + B + C = \pi$, then

(i) $\sin(A+B) = \sin(\pi - C) = \sin C$

(ii) $\cos(A+B) = \cos(\pi - C) = -\cos C$

(iii) $\tan(A+B) = \tan(\pi - C) = -\tan C$

(iv) $\sin(2A+2B) = \sin(2\pi - 2C) = -\sin 2C$

(v) $\cos(2A+2B) = \cos(2\pi - 2C) = \cos 2C$

(vi) $\tan(2A+2B) = \tan(2\pi - 2C) = -\tan 2C$

(vii) $\sin\left(\frac{A+B}{2}\right) = \sin\frac{\pi - C}{2} = \cos\frac{C}{2}$

(viii) $\cos\left(\frac{A+B}{2}\right) = \sin\frac{C}{2}$

(ix) $\tan\left(\frac{A+B}{2}\right) = \cot\frac{C}{2}$

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Illustration 67

$\sin(B + C - A) + \sin(C + A - B) + \sin(A + B - C) = 4\sin A \sin B \sin C$ (where $A + B + C = \pi$)

Solution:

$\sin(\pi - 2A) + \sin(\pi - 2B) + \sin(\pi - 2C) = \sin 2A + \sin 2B + \sin 2C]$

$= 2\sin(A + B)\cos(A - B) + 2\sin C \cos C = 2\sin C[\cos(A - B) + \cos(\pi - A - B)]$

$= 2\sin C[\cos(A - B) - \cos(A + B)] = 4\sin A \sin B \sin C]$

Illustration 68

If $A + B + C = \pi$, Prove that $\sin^2 A + \sin^2 B - \sin^2 C = 2\sin A \sin B \cos C$

Solution:

$\sin^2 A + \sin(B + C)\sin(B - C) = \sin A[\sin(B + C) + \sin(B - C)] = 2\sin A \sin B \cos C]$

Illustration 69

$\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} - \sin^2 \frac{C}{2} = 1 - 2\cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}$ (where $A + B + C = \pi$)

Solution:

$\sin^2 \frac{A}{2} + \sin\left(\frac{B + C}{2}\right)\sin\left(\frac{B - C}{2}\right)$

$= 1 - \cos^2 \frac{A}{2} + \cos \frac{A}{2} \sin\left(\frac{B - C}{2}\right)$

$= 1 - \cos \frac{A}{2}\left[\cos \frac{A}{2} - \sin\left(\frac{B - C}{2}\right)\right] = 1 - \cos \frac{A}{2}\left[\sin\left(\frac{B + C}{2}\right) - \sin\left(\frac{B - C}{2}\right)\right]$

$= 1 - 2\cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}$

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Illustration 70

If A, B, C are the angles of a triangle, then

$\mathbf{\sin 2A + \sin 2B + \sin 2C = 4\sin A \sin B \sin C} \Rightarrow (\sum \sin 2A = 4\prod \sin A)$

Solution:

LHS $= 2\sin(A + B)\cos(A - B) + 2\sin C \cos C = 2\sin C[\cos(A - B) + \cos(\pi - A - B)]$

$= 2\sin C[\cos(A - B) - \cos(A + B)] = 4\sin A \sin B \sin C]$

Illustration 71

$\mathbf{\cos 2A + \cos 2B + \cos 2C = -1 - 4\cos A \cos B \cos C}$ (where $A + B + C = \pi$)

Solution:

$2\cos(A + B)\cos(A - B) + 2\cos^2 C - 1 = 2\cos(\pi - C)\cos(A - B) + 2\cos^2 C - 1]$

$= 2\cos C[\cos C - \cos(A - B)] - 1 = 2\cos C[\cos(\pi - \overline{A + B}) - \cos(A - B)] - 1$

$= 2\cos C[-\cos(A + B) - \cos(A - B)] - 1 = -4\cos A \cos B \cos C - 1$

$\Rightarrow \left(\sum \cos 2A = -1 - 4\prod \cos A\right)$

Illustration 72

$\tan A + \tan B + \tan C = \tan A . \tan B . \tan C$

$\Sigma \tan A = \Pi \tan A$

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Solution:

$\tan(A+B) = \tan(\pi - C) = -\tan C \Rightarrow \frac{\tan A+\tan B}{1-\tan A\tan B} = -\tan C$

$\Rightarrow \tan A + \tan B + \tan C = \tan A.\tan B.\tan C$

If A, B, C are the angles of a triangle i.e. $A + B + C = \pi$, then

(i) $\cot A\cot B + \cot B\cot C + \cot C\cot A = 1$

$\sum \cot A \cot B = 1$

(ii) $\sum \tan\frac{A}{2}\tan\frac{B}{2} = 1$

(iii) $\sum \cot\frac{A}{2} = \Pi\cot\frac{A}{2}$

(iv) $\sin A + \sin B + \sin C = 2\sin\frac{(A+B)}{2}\cos\frac{(A-B)}{2} + 2\sin\frac{C}{2}\cos\frac{C}{2}$

$= 2\cos\frac{C}{2}\left[\cos\frac{(A-B)}{2} + \sin\frac{C}{2}\right]$

$= 2\cos\frac{C}{2}\left[\cos\frac{(A-B)}{2} + \cos\frac{(A+B)}{2}\right]$

$= 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

(v) $\cos A + \cos B + \cos C = 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

Illustration 73

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If $x + y + z = xyz$ prove that $\frac{2x}{1-x^2} + \frac{2y}{1-y^2} + \frac{2z}{1-z^2} = \frac{2x}{1-x^2}\cdot\frac{2y}{1-y^2}\cdot\frac{2z}{1-z^2}$

Solution:

$x = \tan A; y = \tan B; z = \tan C$

Hence $\frac{2\tan A}{1-\tan^2 A} + \frac{2\tan B}{1-\tan^2 B} + \frac{2\tan C}{1-\tan^2 C} = \tan 2A + \tan 2B + \tan 2C$

$\tan A + \tan B + \tan C = \tan A\tan B\tan C$

$\Rightarrow \frac{\tan A+\tan B}{1-\tan A\tan B} = -\tan C \Rightarrow \tan(A + B) = \tan(\pi - C)$

or $A + B = \pi - C \Rightarrow A + B + C = \pi$

$\therefore \quad 2A + 2B = 2\pi - 2C$

$\tan(2A + 2B) = -\tan 2C \Rightarrow \frac{\tan 2A+\tan 2B}{1-\tan 2A\tan 2B} = -\tan 2C$

so $\tan 2A + \tan 2B + \tan 2C = \tan 2A.\tan 2B.\tan 2C = \frac{2\tan A}{1-\tan^2 A}\frac{2\tan B}{1-\tan^2 B}\frac{2\tan C}{1-\tan^2 C}$

$= \frac{2x}{1-x^2}\cdot\frac{2y}{1-y^2}\cdot\frac{2z}{1-z^2}$

Illustration 74

Find the value of $\frac{\sin 50°+\sin 100°+\sin 210°}{\sin 25°\sin 50°\sin 105°}$

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Solution:

Let $A = 25°, B = 50°, C = 105° \Rightarrow A + B + C = \pi$

$$\frac{4\sin 25°\sin 50°\sin 105°}{\sin 25°\sin 50°\sin 105°} = 4$$

Illustration 75

If $A + B + C = 2S$ prove that $\sin(S-A)\sin(S-B) + \sin S\sin(S-C) = \sin A\sin B$.

Solution:

$$\frac{1}{2}[\cos(A-B) - \cos(2S-A-B)] + \frac{1}{2}[\cos C - \cos(2S-C)]$$

$$= \frac{1}{2}[\cos(A-B) - \cos C + \cos C - \cos(A+B)] = \sin A\sin B$$

Illustration 76

In any triangle ABC, $\sin A - \cos B = \cos C$, then angle B is

(A) $\pi/2$ (B) $\pi/3$ (C) $\pi/4$ (D) $\pi/6$

Ans. (A)

Solution:

We have, $\sin A - \cos B = \cos C$

$\sin A = \cos B + \cos C$

$$\Rightarrow 2\sin\frac{A}{2}\cos\frac{A}{2} = 2\cos\left(\frac{B+C}{2}\right)\cos\left(\frac{B-C}{2}\right)$$

$$\Rightarrow 2\sin\frac{A}{2}\cos\frac{A}{2} = 2\cos\left(\frac{\pi-A}{2}\right)\cos\left(\frac{B-C}{2}\right) \because A + B + C = \pi$$

$$\Rightarrow 2\sin\frac{A}{2}\cos\frac{A}{2} = 2\sin\frac{A}{2}\cos\left(\frac{B-C}{2}\right)$$

$\Rightarrow \cos\frac{A}{2} = \cos\frac{B-C}{2}$ or $A = B - C$;But $A + B + C = \pi$

Therefore $2B = \pi \Rightarrow B = \pi/2$

Illustration 77

If $A + B + C = \frac{3\pi}{2}$, then $\cos 2A + \cos 2B + \cos 2C$ is equal to-

(A) $1 - 4\cos A\cos B\cos C$ (B) $4\sin A\sin B\sin C$

(C) $1 + 2\cos A\cos B\cos C$ (D) $1 - 4\sin A\sin B\sin C$

Ans. (D)

Solution:

$\cos 2A + \cos 2B + \cos 2C = 2\cos(A+B)\cos(A-B) + \cos 2C$

$$= 2\cos\left(\frac{3\pi}{2} - C\right)\cos(A-B) + \cos 2C \quad \because A + B + C = \frac{3\pi}{2}$$

$= -2\sin C\cos(A-B) + 1 - 2\sin^2 C = 1 - 2\sin C[\cos(A-B) + \sin C]$

$$= 1 - 2\sin C\left[\cos(A-B) + \sin\left(\frac{3\pi}{2} - (A+B)\right)\right]$$

$= 1 - 2\sin C[\cos(A-B) - \cos(A+B)] = 1 - 4\sin A\sin B\sin C$

Illustration 78

Find $\frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C}$ (where $A + B + C = \pi$)

Solution:

$$\frac{4\sin A\sin B\sin C}{4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}} = 8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

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INEQUALITIES IN A TRIANGLE:

Illustration 79

In any ΔABC show that $\cot^2 A + \cot^2 B + \cot^2 C \geq 1$

Solution:

Let $x = \cot A$, $y = \cot B$ & $z = \cot C$

We know that $\Sigma \cot A \cot B = 1 \Rightarrow \Sigma xy = 1$

Also $(x - y)^2 \geq 0 \quad \Rightarrow x^2 + y^2 \geq 2xy$

$$y^2 + z^2 \geq 2yz$$

$$z^2 + x^2 \geq 2zx$$

$$\therefore \quad 2(x^2 + y^2 + y^2) \geq 2\Sigma xy$$

or $x^2 + y^2 + z^2 \geq 1$

$\Rightarrow \cot^2 A + \cot^2 B + \cot^2 C \geq 1$

APPLICATION OF TRIGONOMETRY IN MAXIMISING & MINIMISING:

Type-I : By property of boundness of trig. function

$\sin x, \cos x \in [-1, 1]$; $-\infty < \tan x, \cot x < \infty$; $\sec x, \operatorname{cosec} x \in (-\infty, -1] \cup [1, \infty)$

$0 \leq \sin^2 x, \cos^2 x \leq 1$; $\sec^2 x, \operatorname{cosec}^2 x \geq 1$; $0 \leq \tan^2 x, \cot^2 x < \infty$

Illustration 80

Find the range

(i) $\sin x, \sin(2x + 3), \sin\frac{x}{2}$

(ii) $4\sin x, 4\cos(3x + 2)$

(iii) $3\sec^2 x$

Solution:

(i) $[-1, 1]$

(ii) $[-4, 4]$

(iii) $[3, \infty]$

Illustration 81

Find range of y given by $y = \cos^4 x - \sin^4 x$

Solution:

$\cos^4 x - \sin^4 x = \cos^2 x - \sin^2 x = \cos 2x \in [-1, 1]$

Illustration 82

$y = 4\tan x.\cos x$

Solution:

$\frac{4\sin x}{\cos x}.\cos x \ \forall \ \cos x \neq 0$

$= 4\sin x \ \forall \ x \neq \frac{\pi}{2}, \frac{3\pi}{2}, \ldots\ldots \in (-4, 4)$

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Illustration 83

$y = \tan^2\left(x - \frac{\pi}{4}\right)$

Solution:

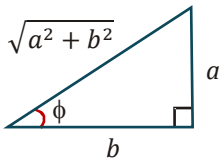
It is square. $\therefore$ minimum = 0

$y \in [0, \infty]$

Type-II : When argument of sine & cosine are same

$$-\sqrt{a^2+b^2} \le a\sin x + b\cos x \le \sqrt{a^2+b^2}$$

Proof : $y = \sqrt{a^2+b^2}\left[\frac{a}{\sqrt{a^2+b^2}}\sin x + \frac{b}{\sqrt{a^2+b^2}}\cos x\right]$



$= \sqrt{a^2+b^2}\,[\sin x \sin\phi + \cos x \cos\phi]$

$\Rightarrow y = \sqrt{a^2+b^2}\cos(x-\phi)$

$y_{max} = \sqrt{a^2+b^2}$; $y_{min} = -\sqrt{a^2+b^2}$

Illustration 84

Range of $y = 3\sin x + 4\cos x + 5$

[**Ans.** $y \in [0, 10]$]

Solution:

$-5 \le 3\sin x + 4\cos x \le 5$

$\Rightarrow -5 + 5 \le 3\sin x + 4\cos x + 5 \le 5 + 5$

$\Rightarrow 0 \le 3\sin x + 4\cos x + 5 \le 10$

Illustration 85

If $b \le 3\sin^2 x + 6\cos^2 x - 4\sin x \cos x + 5 \le a$, find a & b.

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Solution:

Let $E = 3+3\cos^2 x - 2\sin 2x + 5$

$\Rightarrow E = \frac{3}{2}(1+\cos 2x) - 2\sin 2x + 8 \Rightarrow E = \frac{3}{2}\cos 2x - 2\sin 2x + \frac{19}{2}$

$\Rightarrow E_{max} = \sqrt{\frac{9}{4}+4} + \frac{19}{2} = 12$ & $E_{min} = -\sqrt{\frac{9}{4}+4} + \frac{19}{2} = 7$

Illustration 86

Find range of $y = 5\sin\left(x + \frac{\pi}{6}\right) + 3\cos x$

Solution:

$y = 5\left(\frac{\sqrt{3}}{2}\sin x + \frac{1}{2}\cos x\right) + 3\cos x \Rightarrow y = \frac{5\sqrt{3}}{2}\sin x + \frac{11}{2}\cos x$

$y_{max} = \sqrt{\frac{75}{4} + \frac{121}{4}} = \sqrt{\frac{196}{4}} = 7$ & $y_{min} = -7$

Hence range is $[-7, 7]$

Illustration 87

Find maximum & minimum value of $y = \frac{17+(5\sin x + 12\cos x)}{17-(5\sin x + 12\cos x)}$

Solution:

$y_{max} = \frac{17+13}{17-13} = \frac{15}{2}, y_{min} = \frac{17-13}{17+13} = \frac{2}{15}$

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Type-III: Argument of sine & cosine are different or a quadratic in sine/cosine is given then we make a perfect square in sine/cosine & then interpret.

Illustration 88

$y = \cos 2x + 3\sin x$

Solution:

$y = 1 - 2\sin^2 x + 3\sin x = 1 - 2\left[\sin^2 x - \frac{3}{2}\sin x\right]$

$y = 1 - 2\left[\left(\sin x - \frac{3}{4}\right)^2 - \frac{9}{16}\right] = \frac{17}{8} - 2\left(\sin x - \frac{3}{4}\right)^2$

$\Rightarrow y_{min} = \frac{17}{8} - 2\left(-1 - \frac{3}{4}\right)^2 = -\frac{32}{8} = -4$ & $y_{max} = \frac{17}{8} - 2\left(\frac{3}{4} - \frac{3}{4}\right)^2 = \frac{17}{8}$

Illustration 89

Find the range of $y = \sin^2 x - 20\cos x + 1$

Solution:

$y = 1 - \cos^2 x - 20\cos x + 1 \Rightarrow y = 102 - (\cos x + 10)^2$

$\Rightarrow y_{max} = 21$ & $y_{min} = -19$

Illustration 90

Find maximum and minimum values of $y = \cos^2 x - 4\cos x + 13$

Solution:

$y = (\cos x - 2)^2 + 9 \Rightarrow y_{max} = 18$ & $y_{min} = 10$

Type-IV : Making use of reciprocal relationship between tan/cot, sin/cosec, sec/cos.

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Illustration 91

Find minimum value of $y = a^2\tan^2\theta + b^2\cot^2\theta$ $(a, b \geq 0)$

Solution:

Max. value is not meaningful we will take about its minimum value.

$y = (a\tan\theta - b\cot\theta)^2 + 2ab$

This is minimum when $a\tan\theta - b\cot\theta = 0$

$\Rightarrow y = 2ab$, when $a\tan\theta - b\cot\theta = 0$ i.e. $\tan\theta = \sqrt{\frac{b}{a}}$

Note: We can not take $y = (a\tan\theta + b\cot\theta)^2 - 2ab$

$\because$ If $a\tan\theta = -b\cot\theta$

$\Rightarrow \tan^2\theta = \frac{-b}{a}$ Not possible.

Illustration 92

Find minimum value of $y = a^2\sec^2\theta + b^2\operatorname{cosec}^2\theta$

Solution:

$y = a^2 + b^2 + a^2\tan^2\theta + b^2\cot^2\theta = a^2 + b^2 + (a\tan\theta - b\cot\theta)^2 + 2ab$

$y = a^2 + b^2 + 2ab + (a\tan\theta - b\cot\theta)^2$

$\Rightarrow y_{min} = (a + b)^2$, when $\tan\theta = \sqrt{\frac{b}{a}}$

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Illustration 93

$y = 8\sec^2\theta + 18\cos^2\theta$

Solution:

$y = \left(2\sqrt{2}\sec\theta - 3\sqrt{2}\cos\theta\right)^2 + 24$

$\Rightarrow y_{\min} = 24$ when $\cos^2\theta = \frac{2\sqrt{2}}{3\sqrt{2}} = \frac{2}{3}$

Illustration 94

$y = 4\sin^2\theta + \operatorname{cosec}^2\theta$

Solution:

$y = (2\sin\theta - \operatorname{cosec}\theta)^2 + 4$

$\Rightarrow y_{\min} = 4$, when $2\sin\theta = \operatorname{cosec}\theta$ i.e. $\sin^2\theta = \frac{1}{2}$

SUMMATION OF TRIGONOMETRIC SERIES:

Type-I:

Sum of sine/cosine of n angles which are in A.P. i.e. successive arguments of sine or cosine have the same difference.

Illustration 95

Find $S = \sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \ldots\ldots\ldots + \sin(\alpha + \overline{n-1}\beta)$

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Solution:

$2\sin\frac{\beta}{2}.S = 2\sin\frac{\beta}{2}(\sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \ldots\ldots + \sin(\alpha + \overline{n-1}\beta))$

$= \cos(\alpha - \frac{\beta}{2}) - \cos(\alpha + \frac{\beta}{2}) + \cos(\alpha + \frac{\beta}{2}) - \cos(\alpha + \frac{3\beta}{2}) + \ldots\ldots\ldots \cos(\alpha + \overline{n - \frac{3}{2}\beta})$

$- \cos(\alpha + \overline{n - \frac{1}{2}\beta})$

$= \cos(\alpha - \frac{\beta}{2}) - \cos(\alpha + \overline{n - \frac{1}{2}\beta})$

$\Rightarrow S = \sin\frac{n\beta}{2}\sin[\alpha + (n-1)\frac{\beta}{2}] / \sin\frac{\beta}{2} = \frac{\sin\left(\frac{\text{First angle} + \text{Last angle}}{2}\right)\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}}$

Here $\frac{\beta}{2}$ is half of common difference.

Illustration 96

Find $S = \cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \ldots\ldots\ldots + \cos(\alpha + \overline{n-1}\beta)$

Solution:

$2\sin\frac{\beta}{2}.S = 2\sin\frac{\beta}{2}(\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \ldots\ldots\ldots + \cos(\alpha + \overline{n-1}\beta))$

$\Rightarrow S = \sin\frac{n\beta}{2}\cos[\alpha + (n-1)\frac{\beta}{2}] / \sin\frac{\beta}{2}$

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Illustration 97

Find $S = \sin\theta + \sin2\theta + \sin3\theta + \sin4\theta + \ldots\ldots\ldots + \sin n\theta$

Solution:

$2\sin\left(\frac{\theta}{2}\right)S = 2\sin\frac{\theta}{2}[\sin\theta + \sin 2\theta + \sin 3\theta \ldots\ldots\ldots + \sin n\theta]$

$= \cos\frac{\theta}{2} - \cos\frac{3\theta}{2} + \cos\frac{3\theta}{2} - \cos\frac{5\theta}{2} + \cos\frac{5\theta}{2} - \cos\frac{7\theta}{2} \ldots\ldots\ldots + \cos\left(n-\frac{1}{2}\right)\theta - \cos\left(n+\frac{1}{2}\right)\theta$

$\Rightarrow 2\sin\frac{\theta}{2}S = \cos\frac{\theta}{2} - \cos\left(n+\frac{1}{2}\right)\theta$

$\Rightarrow S = \frac{2\sin\left(\frac{n+1}{2}\right)\theta\sin\left(\frac{n\theta}{2}\right)}{2\sin\left(\frac{\theta}{2}\right)}$

or $S = \frac{\sin\left(\frac{\theta+n\theta}{2}\right)\sin\frac{n\theta}{2}}{\sin\theta}$ by standard result.

If $\theta = \frac{2\pi}{n}$ then $S = 0$

Illustration 98

Find $S = \cos\frac{\pi}{19} + \cos\frac{3\pi}{19} + \cos\frac{5\pi}{19} + \ldots\ldots\ldots + \cos\frac{17\pi}{19}$

Solution:

$S = \cos\theta + \cos3\theta + \cos5\theta + \ldots\ldots\ldots + \cos17\theta$, where $\theta = \frac{\pi}{19}$.

$2\sin\theta\, S = 2\sin\theta\,(\cos\theta + \cos 3\theta + \cos 5\theta + \ldots\ldots\ldots + \cos 17\theta)$

$= \sin 2\theta + \sin 4\theta - \sin2\theta + \sin 6\theta - \sin4\theta + \ldots\ldots\ldots + \sin18\theta - \sin 16\theta$

$S = \frac{\sin 18\theta}{2\sin\theta} = \frac{\sin(\pi-\theta)}{2\sin\theta} \qquad \left(\because \theta = \frac{\pi}{19}\right)$

or $S = \frac{1}{2}$

or by standard result

$S = \frac{\cos\left(\frac{\frac{\pi}{19}+\frac{17\pi}{19}}{2}\right)\sin\frac{9\cdot 2\pi}{19\times 2}}{\sin\frac{2\pi}{19\times 2}} = \frac{2\cos\frac{9\pi}{19}\sin\frac{9\pi}{19}}{2\sin\frac{\pi}{19}} = \frac{\sin\frac{18\pi}{19}}{2\sin\frac{\pi}{19}} = \frac{1}{2}$

Illustration 99

Find $S = \sin\theta - \sin2\theta + \sin3\theta - \sin4\theta + \ldots + n$ terms

Solution:

$S = \sin\theta + \sin(\pi + 2\theta) + \sin(2\pi + 3\theta) + \sin(3\pi + 4\theta) + \ldots\ldots + n$ terms

Let $\pi + \theta = d$

$S = \sin\theta + \sin(\theta + d) + \sin(\theta + 2d) + \sin(\theta + 3d) + \ldots\ldots + \sin(\theta + \overline{n-1}d)$

$\Rightarrow S = \frac{\sin(nd/2)\sin(\theta+(n-1)d/2)}{\sin\frac{\theta}{2}}$

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Type II: Splitting the sum of series as difference of 2 terms

Illustration 100

$\operatorname{cosec}x + \operatorname{cosec}2x + \operatorname{cosec}4x + ... + \operatorname{cosec}2^n x = \cot\frac{x}{2} - \cot 2^n x$

Solution:

$$T_1 = \frac{1}{\sin x} = \frac{\sin\frac{x}{2}}{\sin\frac{x}{2}\sin x} = \frac{\sin\left(x-\frac{x}{2}\right)}{\sin\frac{x}{2}\sin x} = \frac{\sin x\cos\frac{x}{2} - \cos x\sin\frac{x}{2}}{\sin\frac{x}{2}\sin x} = \cot\frac{x}{2} - \cot x$$

$$T_2 = \frac{1}{\sin 2x} \times \frac{\sin x}{\sin x} = \frac{\sin(2x-x)}{\sin 2x \sin x} = \cot x - \cot 2x$$

Similarly,

$T_3 = \cot 2x - \cot 2^2 x$, $T_4 = \cot 2^2 x - \cot 2^3 x$,... $T_n = \cot 2^{n-2} x - \cot 2^{n-1} x$, $T_{n+1} = \cot 2^{n-1} x - \cot 2^n x$

$\Rightarrow S = \sum T_n = \cot\frac{x}{2} - \cot 2^n x$

CONTINUED PRODUCT OF SINE & COSINE SERIES:

Σ denotes continued sum & Π denotes continued product

$$\prod_{r=1}^{n} \sin r\theta = \sin\theta \sin 2\theta \sin 3\theta ..... \sin n\theta$$

$$\sum_{r=1}^{n} \sin r\theta = \sin\theta + \sin 2\theta + \sin 3\theta + ... + \sin n\theta$$

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If continued product of cosine series is given such that each angle is double or half of previous angle, then multiply & divide the series by 2 (sine of smallest angle).

Let $P = \cos\theta \cos 2\theta \cos 2^2\theta ..... \cos 2^{n-1}\theta$

multiply Nr & D^r by $\sin\theta$

$$\Rightarrow P = \frac{\sin\theta\cos\theta\cos 2\theta....\cos 2^{n-1}\theta}{\sin\theta} = \frac{2\sin\theta\cos\theta\cos 2\theta.....\cos 2^{n-1}\theta}{2\sin\theta} = \frac{\sin 2\theta\cos 2\theta.....\cos 2^{n-1}\theta}{2\sin\theta}$$

$$= \frac{2\sin 2\theta\cos 2\theta.....\cos 2^{n-1}\theta}{2^2\sin\theta} = \frac{\sin 2^n\theta}{2^n\sin\theta}$$

Illustration 101

Prove that $\cos 20\cos 40\cos 80 = \frac{1}{8}$ or $\cos\frac{\pi}{9}\cos\frac{2\pi}{9}\cos\frac{4\pi}{9} = \frac{1}{8}$

Solution:

$$\frac{2\sin 20}{2\sin 20} \times \cos 20\cos 40\cos 80 = \frac{2\sin 40\cdot\cos 40\cos 80}{2^2\sin 20} = \frac{2\sin 80\cos 80}{2^3\sin 20} = \frac{\sin 160}{2^2\sin 20} = \frac{\sin 20}{2^3\sin 20} = \frac{1}{8}$$

Direct: $= \frac{\sin 2^3(20)}{2^3\sin(20)} = \frac{\sin 160}{8\sin 20} = \frac{1}{8}$

Illustration 102

$$\cos\frac{2\pi}{15}\cos\frac{4\pi}{15}\cos\frac{8\pi}{15}\cos\frac{14\pi}{15} = -\frac{1}{16}$$

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Solution:

$$\cos\left(\frac{2\pi}{15}\right)\cos\left(\frac{4\pi}{15}\right)\cos\left(\frac{8\pi}{15}\right)\cos\left(\pi-\frac{\pi}{15}\right) = -\cos\frac{\pi}{15}\cos\frac{2\pi}{15}\cos\frac{4\pi}{15}\cos\frac{8\pi}{15}$$

Multiply & divide by $2\sin\left(\frac{\pi}{15}\right)$ & solve

Direct method:

$$= -\frac{\sin 2^4\left(\frac{\pi}{15}\right)}{2^4\sin\frac{\pi}{15}} = -\frac{\sin\frac{16\pi}{15}}{16\cdot\sin\frac{\pi}{15}} = -\frac{\sin\left(\pi+\frac{\pi}{15}\right)}{16\cdot\sin\frac{\pi}{15}} = \frac{1}{16}$$

Illustration 103

$$P = \sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}\sin\frac{7\pi}{14}\sin\frac{9\pi}{14}\sin\frac{11\pi}{14}\sin\frac{13\pi}{14}$$

Solution:

$$P = \sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}\times 1\times\sin\left(\pi-\frac{5\pi}{14}\right)\sin\left(\pi-\frac{3\pi}{14}\right)\sin\left(\pi-\frac{\pi}{14}\right)$$

$$P = \left(\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}\right)^2 = q^2 \text{ (let)}$$

$$\Rightarrow q = \sin\left(\frac{7\pi-6\pi}{14}\right)\sin\left(\frac{7\pi-4\pi}{14}\right)\sin\left(\frac{7\pi-2\pi}{14}\right)$$

$$\text{or } q = \cos\frac{3\pi}{7}\cos\frac{2\pi}{7}\cos\frac{\pi}{7}$$

$$\Rightarrow q = -\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7} = -\frac{\sin\frac{8\pi}{7}}{8\sin\frac{\pi}{7}} = -\frac{\sin\left(\pi+\frac{\pi}{7}\right)}{8\sin\frac{\pi}{7}} = \frac{1}{8}. \qquad \therefore P = \frac{1}{64}$$

Illustration 104

Find the value of $\sin\frac{\pi}{16}\sin\frac{3\pi}{16}\sin\frac{5\pi}{16}\sin\frac{7\pi}{16}$

Solution:

$$\sin\frac{\pi}{16}\sin\frac{3\pi}{16}\cos\frac{3\pi}{16}\cos\frac{\pi}{16} = \frac{1}{4}\left[\sin\frac{2\pi}{16}\sin\frac{6\pi}{16}\right] = \frac{1}{8}\left[2\sin\frac{\pi}{8}\sin\frac{3\pi}{8}\right]$$

$$= \frac{1}{8}\left(2\sin\frac{\pi}{8}\cos\frac{\pi}{8}\right) = \frac{1}{8}\sin\frac{\pi}{4} = \frac{1}{8\sqrt{2}}$$

Illustration 105

$$\cos\frac{\pi}{9}\cos\frac{2\pi}{9}\cos\frac{3\pi}{9}\cos\frac{4\pi}{9} = \frac{1}{16}$$

Solution:

$$= \frac{1}{2}\left[\cos\frac{\pi}{9}\cdot\cos\frac{2\pi}{9}\cdot\cos\frac{4\pi}{9}\right] = \left(\frac{1}{2}\right)\frac{\sin 2^3\frac{\pi}{9}}{2^3\sin\frac{\pi}{9}} = \frac{1}{16}\frac{\sin\frac{8\pi}{9}}{\sin\frac{\pi}{9}} = \frac{1}{16}$$

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Illustration 106

Show that $2^{\sin x}+2^{\cos x}\geq 2^{1-\frac{1}{\sqrt{2}}}$

Solution:

Since A.M. of two positive quantities ≥ Their G.M. we have

$$\frac{2^{\sin x}+2^{\cos x}}{2}\geq\sqrt{2^{\sin x}2^{\cos x}}=\sqrt{2^{\sin x}2^{\cos x}}=\sqrt{2^{\sqrt{2}\sin\left(x+\frac{\pi}{4}\right)}}\geq\sqrt{2^{-\sqrt{2}}}$$

$$\Rightarrow 2^{\sin x}+2^{\cos x}\geq 2\times 2^{-\frac{1}{\sqrt{2}}}=2^{1-\frac{1}{\sqrt{2}}}$$

Illustration 107

If $\tan 6\theta = p/q$, find the value of $\frac{1}{2}(p\operatorname{cosec}2\theta - q\sec 2\theta)$ in terms of p and q.

Solution:

Here, we have $\tan 6\theta = p/q$

$$\Rightarrow \frac{\sin 6\theta}{\cos 6\theta}=\frac{p}{q}$$

$$\Rightarrow \frac{p}{\sin 6\theta}=\frac{q}{\cos 6\theta}=\frac{\sqrt{p^2+q^2}}{\sqrt{1}}=\sqrt{p^2+q^2}$$

Now $y=\frac{1}{2}(p\operatorname{cosec}2\theta - q\sec 2\theta)=\frac{1}{2}\left(\frac{p}{\sin 2\theta}-\frac{q}{\cos 2\theta}\right)$

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$$\Rightarrow y=\frac{1}{2}\left[\frac{p\cos 2\theta - q\sin 2\theta}{\sin 2\theta\cos 2\theta}\right]$$

$$=\left[\frac{2k\sin 6\theta\cos 2\theta - 2k\cos 6\theta\sin 2\theta}{4\sin 2\theta\cos 2\theta}\right]$$

$$=k\frac{\sin(6\theta-2\theta)}{\sin 4\theta}=k=\sqrt{p^2+q^2}$$

Illustration 108

The maximum value of $y=\frac{1}{\sin^6 x+\cos^6 x}$ is_____

Solution:

$$\sin^6 x+\cos^6 x=(\sin^2 x+\cos^2 x)(\sin^4 x+\cos^4 x-\sin^2 x\cos^2 x)$$

$$1-3\sin^2 x\cos^2 x=1-\frac{3(\sin 2x)^2}{4}$$

$$\Rightarrow y=\frac{4}{4-3(\sin 2x)^2}$$

$$\Rightarrow y_{max}=\frac{4}{4-3(1)}=4$$

Illustration 109

The value of $\operatorname{cosec}10°+\operatorname{cosec}50°-\operatorname{cosec}70°$ is _____

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Solution:

$$\frac{1}{\sin 10^\circ}+\frac{1}{\sin 50^\circ}-\frac{1}{\sin 70^\circ}=\frac{1}{\cos 80^\circ}+\frac{1}{\cos 40^\circ}-\frac{1}{\cos 20^\circ}$$
$$=\frac{\cos 40^\circ\cos 20^\circ+\cos 80^\circ\cos 20^\circ-\cos 40^\circ\cos 80^\circ}{\cos 20^\circ\cos 40^\circ\cos 80^\circ}$$
$$=8[\cos 20^\circ(\cos 40^\circ+\cos 80^\circ)-\cos 40^\circ\cos 80^\circ]$$
$$=8[2\cos 20^\circ\cos 60^\circ\cos 20^\circ-\cos 40^\circ\cos 80^\circ]$$
$$=4[2\cos^2 20^\circ-2\cos 40^\circ\cos 80^\circ]$$
$$=4[1+\cos 40^\circ-(\cos 120^\circ+\cos 40^\circ)]$$
$$=4\times\frac{3}{2}=6$$

Illustration 110

If $\log_{10}\sin x+\log_{10}\cos x=-1$ and $\log_{10}(\sin x+\cos x)=\frac{(\log_{10}n)-1}{2}$. Then the value of $\frac{n}{3}$ is_____

Solution:

Given $\log_{10}\left(\frac{\sin 2x}{2}\right)=-1$

$$\Rightarrow\frac{\sin 2x}{2}=\frac{1}{10}$$
$$\Rightarrow\sin 2x=\frac{1}{5}$$

Also $\log_{10}(\sin x+\cos x)=\frac{\log_{10}\left(\frac{n}{10}\right)}{2}$

$$\Rightarrow\log_{10}(\sin x+\cos x)^2=\log_{10}\left(\frac{n}{10}\right)$$
$$\Rightarrow 1+\sin 2x=\frac{n}{10}$$
$$\Rightarrow 1+\frac{1}{5}=\frac{n}{10}$$
$$\Rightarrow\frac{6}{5}=\frac{n}{10}$$
$$\Rightarrow\frac{n}{3}=4$$

Illustration 111

If $\sin^3 x\cos 3x+\cos^3 x\sin 3x=\frac{3}{8}$, Then the value of $8\sin 4x$ is _____

Solution:

$$4\sin^3 x\cos 3x+4\cos^3 x\sin 3x=\frac{3}{2}$$
$$\Rightarrow(3\sin x-\sin 3x)\cos 3x+(3\cos x+\cos 3x)\sin 3x=\frac{3}{2}$$
$$\Rightarrow 3[\sin x\cos 3x+\cos x\sin 3x]=\frac{3}{2}$$
$$\Rightarrow\sin 4x=\frac{1}{2}$$
$$\Rightarrow 8\sin 4x=4$$

Illustration 112

If $\cos\theta-\sin\theta=\frac{1}{5}$ where $0<\theta<\frac{\pi}{2}$

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Column I	Column II
a. $(\cos\theta + \sin\theta)/2$	**p.** $\frac{4}{5}$
b. $\sin 2\theta$	**q.** $\frac{7}{10}$
c. $\cos 2\theta$	**r.** $\frac{24}{25}$
d. $\cos\theta$	**s.** $\frac{7}{25}$

Ans. $[a \to q; b \to r; c \to s; d \to p]$

Solution:

$\cos\theta - \sin\theta = \frac{1}{5}$ Where $0 < \theta < \frac{\pi}{2}$ ...(i)

Squaring both side of Eq. (i), we get

$1 - \sin 2\theta = \frac{1}{25}$

$\Rightarrow \sin 2\theta = \frac{24}{25} \Rightarrow \cos 2\theta = \frac{7}{25}$

Also , $(\cos\theta + \sin\theta)^2 = (\cos\theta - \sin\theta)^2 + 4\cos\theta\sin\theta = \frac{1}{25} + 2\sin 2\theta = \frac{1}{25} + \frac{48}{25} = \frac{49}{25}$

$\Rightarrow \cos\theta + \sin\theta = \frac{7}{5}$ ...(ii)

$\Rightarrow (\cos\theta + \sin\theta)/2 = \frac{7}{10}$

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Also solving Eqs. (i) and (ii), we get $\cos\theta = \frac{4}{5}$

Illustration 113

For all real values of θ

Column I	Column II
a. $A = \sin^2\theta + \cos^4\theta$	**p.** $A \in [-1,1]$
b. $A = 3\cos^2\theta + \sin^4\theta$	**q.** $A \in \left[\frac{3}{4}, 1\right]$
c. $A = \sin^2\theta - \cos^4\theta$	**r.** $A \in [2\sqrt{2}, \infty]$
d. $A = \tan^2\theta + 2\cot^2\theta$	**s.** $A \in [1,3]$

Ans. $[a \to q;\ b \to s;\ c \to p;\ d \to r]$

Solution:

a. $A = \sin^2\theta + \cos^4\theta$

$= \frac{1 - \cos 2\theta}{2} + \left(\frac{1 + \cos 2\theta}{2}\right)^2$

$= \frac{1}{2} - \frac{1}{2}\cos 2\theta + \frac{1}{4} + \frac{1}{2}\cos 2\theta + \frac{1}{4}\cos^2 2\theta$

$= \frac{3}{4} + \frac{1}{4}\left(\frac{\cos 4\theta + 1}{2}\right) = \frac{3}{4} + \frac{1}{8} + \frac{1}{8}\cos 4\theta$

Now $-1 \le \cos\theta \le 1$

$\frac{-1}{8} \le \frac{\cos 4\theta}{8} \le \frac{1}{8}$

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$\Rightarrow \frac{3}{4}+\frac{1}{8}-\frac{1}{8} \le \frac{3}{4}+\frac{1}{4}\left(\frac{1+\cos 4\theta}{2}\right) \le \frac{3}{4}+\frac{1}{8}+\frac{1}{8}$

$\Rightarrow \frac{3}{4} \le A \le 1$

b. $A = 3\cos^2\theta + \sin^4\theta = 3\frac{1+\cos 2\theta}{2} + \left(\frac{1-\cos 2\theta}{2}\right)^2$

$= \frac{3+3\cos 2\theta}{2} + \frac{1-2\cos 2\theta+\cos^2 2\theta}{4}$

$= \frac{7+4\cos 2\theta+\cos^2 2\theta}{4} = \frac{(\cos 2\theta+2)^2+3}{4}$

Now $1 \le \cos 2\theta + 2 \le 3$

$\Rightarrow 1 \le \frac{(\cos 2\theta+2)^2+3}{4} \le 3$

c. $A = \sin^2\theta - \cos^4\theta$

$= \frac{1-\cos 2\theta}{2} - \left(\frac{1+\cos 2\theta}{2}\right)^2$

$= \frac{1}{2} - \frac{1}{2}\cos 2\theta - \frac{1}{4} - \frac{1}{2}\cos 2\theta - \frac{1}{4}\cos^2 2\theta$

$= \frac{1}{4} - \cos 2\theta - \frac{1}{4}\cos^2 2\theta$

$= -\left(\frac{1}{4}\cos^2 2\theta + \cos 2\theta - \frac{1}{4}\right)$

$= \frac{5}{4} - \left(\frac{1}{2}\cos 2\theta + 1\right)^2$

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Now $-\frac{1}{2} \le \frac{1}{2}\cos 2\theta \le \frac{1}{2}$

$\Rightarrow \frac{1}{2} \le \frac{1}{2}\cos 2\theta + 1 \le \frac{3}{2}$

$\Rightarrow \frac{1}{4} \le \left(\frac{1}{2}\cos 2\theta + 1\right)^2 \le \frac{9}{4}$

$\Rightarrow -\frac{9}{4} \le -\left(\frac{1}{2}\cos 2\theta + 1\right)^2 \le -\frac{1}{4}$

$\Rightarrow -1 \le \frac{5}{4} - \left(\frac{1}{2}\cos 2\theta + 1\right)^2 \le 1$

d. $\tan^2\theta + 2\cot^2\theta = \left(\tan\theta - \sqrt{2}\cot\theta\right)^2 + 2\sqrt{2} \ge 2\sqrt{2}$

Illustration 114

If $\cos\alpha + \cos\beta = 1/2$ and $\sin\alpha + \sin\beta = 1/3$

Column I	Column II
a. $\cos\left(\frac{\alpha+\beta}{2}\right)$	**p.** $\pm\frac{\sqrt{13}}{12}$
b. $\cos\left(\frac{\alpha-\beta}{2}\right)$	**q.** $\frac{2}{3}$
c. $\tan\left(\frac{\alpha+\beta}{2}\right)$	**r.** $\pm\frac{3}{\sqrt{13}}$
d. $\tan\left(\frac{\alpha-\beta}{2}\right)$	**s.** $\pm\sqrt{\frac{131}{13}}$

Ans. $[a \to r; b \to p; c \to q; d \to s]$

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Solution:

$\cos\alpha+\cos\beta=\frac{1}{2}$

$\Rightarrow 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)=\frac{1}{2}$ ...(i)

$\sin\alpha+\sin\beta=\frac{1}{3}$

$\Rightarrow 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)=\frac{1}{3}$ ...(ii)

Dividing Eq. (ii) by Eq.(i). we get

$\tan\left(\frac{\alpha+\beta}{2}\right)=\frac{2}{3}$

$\Rightarrow \cos\left(\frac{\alpha+\beta}{2}\right)=\pm\frac{3}{\sqrt{13}}$

Squaring and adding the given result, we have

$2+2\cos(\alpha-\beta)=\frac{13}{36}$

$\Rightarrow \cos(\alpha-\beta)=-\frac{59}{72}$

$\Rightarrow 2\cos^2\left(\frac{\alpha-\beta}{2}\right)=1-\frac{59}{72}=\frac{13}{72}$

$\Rightarrow \cos\left(\frac{\alpha-\beta}{2}\right)=\pm\frac{\sqrt{13}}{12}$

$\Rightarrow \tan\left(\frac{\alpha-\beta}{2}\right)=\pm\sqrt{\frac{131}{13}}$

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Illustration 115

Column-I	Column-II
(a) $\cos20°+\cos80°-\sqrt{3}\cos50°$	**(p)** -1
(b) $\cos0°+\cos\frac{\pi}{7}+\cos\frac{2\pi}{7}+\cos\frac{3\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{5\pi}{7}+\cos\frac{6\pi}{7}$	**(q)** $-\frac{3}{4}$
(c) $\cos20°+\cos40°+\cos60°-4\cos10°\cos20°\cos30°$	**(r)** 1
(d) $\cos20°\cos100°+\cos100°\cos140°-\cos140°\cos200°$	**(s)** 0

Ans. $[a\to s; b\to r; c\to p; d\to q]$

Solution:

(a) $\cos20°+\cos80°-\sqrt{3}\cos50°=2\cos30°\cos50°-\sqrt{3}\cos50°$

$=\sqrt{3}\cos50°-\sqrt{3}\cos50°=0$

(b) $\cos0°+\cos\frac{\pi}{7}+\cos\frac{2\pi}{7}+\cos\frac{3\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{5\pi}{7}+\cos\frac{6\pi}{7}$

$=1+\left(\cos\frac{\pi}{7}+\cos\frac{6\pi}{7}\right)+\left(\cos\frac{2\pi}{7}+\cos\frac{5\pi}{7}\right)+\left(\cos\frac{3\pi}{7}+\cos\frac{4\pi}{7}\right)$

$=1+\left(\cos\frac{\pi}{7}+\cos\left(\pi-\frac{\pi}{7}\right)\right)+\left(\cos\frac{2\pi}{7}+\cos\left(\pi-\frac{2\pi}{7}\right)\right)+\left(\cos\frac{3\pi}{7}+\cos\left(\pi-\frac{3\pi}{7}\right)\right)$

$=1+0+0+0=1$

(c) $\cos20°+\cos40°+\cos60°-4\cos10°\cos20°\cos30°$

$=2\cos30°\cos10°+2\cos^2 30°-1-4\cos10°\cos20°\cos30°$

$=2\cos30°\,(2\cos10°\cos20°)-1-4\cos10°\cos20°\cos30°=-1$

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(d) $\cos20°\cos100°+\cos100°\cos140°-\cos140°\cos200°$

$=\frac{1}{2}(\cos120°+\cos80°+\cos240°+\cos40°-\cos340°-\cos60°)$

$=\frac{1}{2}\left(-\frac{1}{2}+\cos80°-\frac{1}{2}+\cos40°-\cos340°-\frac{1}{2}\right)=\frac{1}{2}\left(-\frac{3}{2}+\cos80°+\cos40°-\cos20°\right)$

$=\frac{1}{2}\left(-\frac{3}{2}+2\cos60°\cos20°-\cos20°\right)=\frac{1}{2}\left(-\frac{3}{2}\right)=-\frac{3}{4}$

Illustration 116

If $\sin\alpha = A\sin(\alpha+\beta)$, $A\neq 0$, then the value of $\tan\alpha$ is

(A) $\frac{A\sin\beta}{1-A\cos\beta}$ (B) $\frac{A\sin\beta}{1+A\cos\beta}$ (C) $\frac{A\cos\beta}{1-A\sin\beta}$ (D) $\frac{A\cos\beta}{1+A\sin\beta}$

Ans. (A)

Solution:

$\sin\alpha = A\sin(\alpha+\beta)=A(\sin\alpha\cos\beta+\sin\beta\cos\alpha)$

$\Rightarrow \sin\alpha(1-A\cos\beta)=A\sin\beta\cos\alpha$ ...(i)

$\Rightarrow \tan\alpha=\frac{A\sin\beta}{(1-A\cos\beta)}$ ...(ii)

Illustration 117

If $\sin\alpha = A\sin(\alpha+\beta)$, $A\neq 0$, the value of $\tan\beta$ is

(A) $\frac{\sin\alpha(1+A\cos\beta)}{A\cos\alpha\cos\beta}$ (B) $\frac{\sin\alpha(1-A\cos\beta)}{A\cos\alpha\cos\beta}$ (C) $\frac{\cos\alpha(1-A\sin\beta)}{A\cos\alpha\cos\beta}$ (D) $\frac{\cos\alpha(1+A\sin\beta)}{A\cos\alpha\cos\beta}$

Ans. (B)

Solution:

$\tan\beta=\frac{\sin\beta}{\cos\beta}=\frac{(1-A\cos\beta)\tan\alpha}{A\cos\beta}=\frac{(1-A\cos\beta)\sin\alpha}{A\cos\alpha\cos\beta}$ [from Eqs. (i) and (ii)]

Illustration 118

Column-I	Column-II
(a) Suppose ABC is a triangle with three acute angle A, B and C. The point whose coordinates are $(\cos B-\sin A, \sin B-\cos A)$ can be in the	(p) Ist quadrant
(b) If $2^{\sin\theta}>1$ and $3^{\cos\theta}<1$, then $\theta\in$	(q) 2nd quadrant
(c) $\lvert\cos x+\sin x\rvert=\lvert\sin x\rvert+\lvert\cos x\rvert$	(r) 3rd quadrant
(d) If $\sqrt{\frac{1-\sin A}{1+\sin A}}+\frac{\sin A}{\cos A}=\frac{1}{\cos A}$, for all permission values of A, then A can belong to	(s) 4th quadrant

Ans. $[a\to q; b\to q; c\to p,r; d\to p,s]$

Solution:

(a) Since angle A, B and C are acute angle

$\therefore A+B>\frac{\pi}{2}$

$A>\frac{\pi}{2}-B$

$\sin A-\cos B>0$

$\Rightarrow \cos B-\sin A<0$ ...(i)

Again $B>\frac{\pi}{2}-A$

$\sin B>\cos A$

$\sin B-\cos A>0$ ...(ii)

from Eq. (i) and (ii), we get that x-coordinates is –ve and y-coordinates is +ve.

Therefore, point lies in 2nd quadrant only

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(b) $2^{\sin\theta} > 1 \Rightarrow \sin\theta > 0 \Rightarrow \theta \in 1^{st}$ or 2nd quadrant

$3^{\cos\theta} < 1 \Rightarrow \cos\theta < 0 \Rightarrow \theta \in 2^{nd}$ or 3rd quadrant

Hence $\theta \in 2^{nd}$ quadrant

(c) $|\cos x + \sin x| = |\sin x| + |\cos x|$

$\Rightarrow \cos x$ and $\sin x$ must have same sign or at least one in zero

$\Rightarrow x \in 1^{st}$ or 3rd quadrant

(d) L.H.S $= \frac{1-\sin A}{|\cos A|} + \frac{\sin A}{\cos A} = \frac{1}{\cos A}$ which is truly only if $|\cos A| = \cos A$

$\Rightarrow A \in 1^{st}$ or 4th quadrant

Illustration 119

Column-I	Column-II
(a) If $x^2 + y^2 = 1$ and $p = (3x - 4x^3)^2 + (3y - 4y^3)^2$, then p is equal to	**(p)** 1
(b) If $a + b = 3 - \cos 4\theta$ and $a - b = 4\sin 2\theta$, then the maximum value of $(ab)$is	**(q)** 4
(c) The least positive integral value of x for which $3\cos\theta = x^2 - 8x + 19$	**(r)** 5
(d) If $x = \frac{4\lambda}{1+\lambda^2}$ and $y = \frac{2-2\lambda^2}{1+\lambda^2}$, where λ is a real parameter, then $x^2 - xy + y^2$ lies between $[a, b]$ then $(a + b)$ is	**(s)** 8

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Ans. $[\boldsymbol{a \to p; b \to p; c \to q; d \to s}]$

Solution:

(a) $x = \sin\theta, y = \cos\theta$

$p = (3\sin\theta - 4\sin^3\theta)^2 + (3\cos\theta - 4\cos^3\theta)^2 = \sin^2 3\theta + \cos^2 3\theta = 1$

(b) on adding, we get $a = \frac{3-\cos 4\theta + 4\sin 2\theta}{2} = (1 + \sin 2\theta)^2$

on subtracting, we get $b = (1 - \sin 2\theta)^2 \Rightarrow ab = \cos^4 2\theta \leq 1$

(c) $3\cos\theta = x^2 - 8x + 19$

$\Rightarrow 3\cos\theta = (x - 4)^2 + 3$

Now L.H.S. $= 3\cos\theta \leq 3$ or L.H.S. has the greatest value 3.

But R.H.S. $(x - 4)^2 + 3 \geq 3$ OR R.H.S. has the least value 3.

Hence, L.H.S. = R.H.S. when $3\cos\theta = (x - 4)^2 + 3 = 3$

$\Rightarrow \cos\theta = 1$ and $x - 4 = 0$

$\Rightarrow \theta = 2n\pi$ and $x = 4$, where $n \in Z$.

(d) $\lambda = \tan\theta$

$x = 2\sin 2\theta$ and $y = 2\cos 2\theta$

$E = x^2 - xy + y^2 = 4 - 4\sin 2\theta\cos 2\theta = 4 - 2\sin 4\theta$

$E \in [2,6] \Rightarrow a + b = 8$

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Illustration 120

	Column-I	Column-II
(a)	In triangle ABC, $3\sin A + 4\cos B = 6$ $3\cos A + 4\sin B = 1$, then $\angle C$ can be	**(p)** 60°
(b)	In any triangle, if $(\sin A + \sin B + \sin C)$ $(\sin A + \sin B - \sin C) = 3\sin A \sin B$, then the angle C	**(q)** 30°
(c)	If $8\sin x \cos^5 x - 8\sin^5 x \cos x = 1$, then $x =$	**(r)** 165°
(d)	'O' is the center of inscribed circle in a 30°– 60°– 90° triangle ABC with right angled at C. if the circle is tangent to AB at D, then the angle $\angle COD$ is	**(s)** 7.5°

Ans. $[a \to q; b \to p;\ c \to s; d \to r]$

Solution:

(a) $9 + 16 + 24\sin(A + B) = 37$ (on squaring and adding)

$24\sin(A + B) = 12$

$\sin(A + B) = \frac{1}{2} \Rightarrow \sin C = \frac{1}{2}$

$C = 30°$ or $150°$

$\Rightarrow C = 30°$

(b) $(\sin A + \sin B)^2 - \sin^2 C = 3\sin A \sin B$

$\Rightarrow \sin^2 A - \sin^2 C + \sin^2 B = \sin A \sin B$

$\Rightarrow \sin(A + C)\sin(A - C) + \sin^2 B = \sin A \sin B$

$\Rightarrow \sin B[\sin(A - C) + \sin(A + C)] = \sin A \sin B$

$\Rightarrow 2\sin A \cos C = \sin A$ (as $\sin B \neq 0$)

$\Rightarrow \cos C = \frac{1}{2}$

$\Rightarrow C = 60°$

(c) $2\sin x \cos x\,[4\cos^4 x - 4\sin^4 x] = 1$

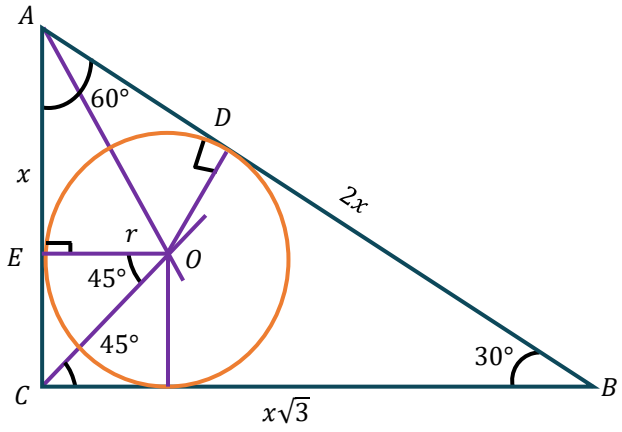
$\Rightarrow (\sin 2x)\,[2(\cos^2 x + \sin^2 x)][2(\cos^2 x - \sin^2 x)] = 1$

$\Rightarrow (\sin 2x)2\times 2\cos 2x = 1$

$\Rightarrow 2\sin 4x = 1$

$\Rightarrow \sin 4x = \frac{1}{2} \Rightarrow 4x = 30° \Rightarrow x = 7.5°$

(d)



Obviously $AEOD$ is a cyclic quadrilateral we have

$\angle COD = 120° + 45° = 165°$

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